# Evolutionary Multitasking for Feature Selection in High-Dimensional Classification via Particle Swarm Optimization

Ke Chen, *Member, IEEE*, Bing Xue, *Senior Member, IEEE*, Mengjie Zhang, *Fellow, IEEE*, and Fengyu Zhou

***Abstract*—Feature selection (FS) is an important preprocessing technique for improving the quality of feature sets in many practical applications. Particle swarm optimization (PSO) has been widely used for FS due to being efficient and easy to implement. However, when dealing with high-dimensional data, most of the existing PSO-based FS approaches face the problems of falling into local optima and high-computational cost. Evolutionary multitasking is an effective paradigm to enhance global search capability and accelerate convergence by knowledge transfer among related tasks. Inspired by evolutionary multitasking, this article proposes a multitasking PSO approach for high-dimensional FS. The approach converts a high-dimensional FS task into several related low-dimensional FS tasks, then finds an optimal feature subset by knowledge transfer between these low-dimensional FS tasks. Specifically, a novel task generation strategy based on the importance of features is developed, which can generate highly related tasks from a dataset adaptively. In addition, a new knowledge transfer mechanism is presented, which can effectively implement positive knowledge transfer among related tasks. The results demonstrate that the proposed method can evolve a feature subset with higher classification accuracy in a shorter time than other state-of-the-art FS methods on high-dimensional classification.**

***Index Terms*—Evolutionary multitasking, feature selection (FS), high-dimensional classification, knowledge transfer, particle swarm optimization (PSO).**

Manuscript received June 18, 2020; revised October 29, 2020, January 31, 2021, and May 9, 2021; accepted July 2, 2021. Date of publication July 26, 2021; date of current version May 30, 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2017YFB1302400; in part by the National Natural Science Foundation of China under Grant 61773242, Grant 61803227, Grant 61922072, and Grant 61876169; in part by the Major Agricultural Applied Technological Innovation Projects of Shandong Province under Grant SD2019NJ014; in part by the Shandong Natural Science Foundation under Grant ZR2019MF064; in part by the Intelligent Robot and System Innovation Center Foundation under Grant 2019IRS19; in part by the Marsden Fund of New Zealand Government under Grant VUW1913 and Grant VUW1914; in part by the Science for Technological Innovation Challenge (SfTI) Fund under Grant E3603/2903; in part by the University Research Fund at Victoria University of Wellington under Grant 223805/3986; and in part by the MBIE Data Science SSIF Fund under Grant RTVU1914. *(Corresponding author: Fengyu Zhou.)*

Ke Chen is with the School of Electrical Engineering, Zhengzhou University, Zhengzhou 450001, China (e-mail: chenkezixf@163.com).

Bing Xue and Mengjie Zhang are with the School of Engineering and Computer Science, Victoria University of Wellington, Wellington 6140, New Zealand (e-mail: bing.xue@ecs.vuw.ac.nz; mengjie.zhang@ecs.vuw.ac.nz).

Fengyu Zhou is with the School of Control Science and Engineering, Shandong University, Jinan 250061, China (e-mail: fyzhou_sdu@163.com).

This article has supplementary material provided by the authors and color versions of one or more figures available at https://doi.org/10.1109/TEVC.2021.3100056.

Digital Object Identifier 10.1109/TEVC.2021.3100056

## I. Introduction

CLASSIFICATION is a widely-studied data mining task in academia and industry, which aims to classify unseen data based on the information presented by its features [1]. With the rapid development of emerging techniques, high-dimensional data become common in many real-world applications, such as job shop scheduling [2] and text classification [3]. In such data, many features (i.e., irrelevant and redundant features) are not useful for class prediction, and they not only increase the computational complexity of a learning algorithm but also degrade the classification performance due to "the curse of dimensionality" [4]. Feature selection (FS) is an effective data preprocessing technique, and is capable of solving this challenge by choosing a subset of relevant features from the original features [5].

Generally, existing FS approaches can be grouped into three categories according to different evaluation criteria, i.e., filter-based approaches, wrapper-based approaches, and embedded-based approaches [6]. Filter-based approaches use data intrinsic measures (i.e., distance, correlation, consistency, information gain) to evaluate and determine selected features [7]. Filter-based approaches typically require less computational resources, but often achieve lower classification accuracy than wrapper-based approaches. Wrapper-based approaches utilise a learning algorithm to evaluate the selected features [8]. Since the feature subset is determined by a learning algorithm, wrapper-based approaches can typically achieve higher classification accuracy than filter-based approaches. However, wrapper-based approaches are generally computationally more expensive than filter-based approaches because they need more computational resources to assess each selected feature subset [9]. Embedded-based approaches incorporate FS into the model training process and are associated with a specific classification algorithm. In general, wrapper-based and embedded-based approaches can usually obtain better classification accuracy than filter-based approaches, but with the price of higher computational cost and less generalization [10].

FS is a combinatorial optimization problem [11], and has $2^n$ possible feature combinations, where $n$ denotes the number of all available features in the dataset. Therefore, it is almost impossible to conduct an exhaustive search to find the best feature subset, especially when the number of features

1089-778X © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

is large. A large number of heuristic search approaches (e.g., sequential forward or backward selection, sequential forward or backward floating selection) have been applied to tackle this problem [10], [12]. Unfortunately, the primary deficiency of these approaches is that they often achieve local optimal solutions during the FS process.

Recently, FS approaches based on evolutionary computation (EC) have received widespread attention due to their ability in finding satisfactory solutions within a reasonable time via global search strategies. Particle swarm optimization (PSO) [13] is an EC algorithm based on swarm intelligence. Compared with many other EC algorithms, PSO has the advantages of being efficient, easy to implement, and having fewer parameters to adjust [14]. In addition, PSO has been widely applied for solving FS problems and its effectiveness has been verified [15]–[17]. Therefore, PSO is adopted as an effective search technique in this study. To date, many PSO-based FS approaches have been developed in [11], [18], [19]. However, owing to the rapid growth of the number of features, FS in high-dimensional problems brings new challenges for PSO. For instance, a PSO-based unsupervised FS was developed in [11], where two strategies (i.e., the space reduction strategy and the local filter search strategy) were introduced to PSO to choose a feature subset. However, since interactions between features are ignored during the search process, this method still faces the problem of falling into local optima. In [18], a PSO-based multiobjective FS method was proposed, where three measures (i.e., the fuzzy dominance relationship, the fuzzy crowding distance, and the tolerance coefficient) were applied to determine the Pareto-optimal feature subsets. However, since the calculation of these three measures requires more computing resources, this method still faces the problem of high-computational cost, especially the number of features is large. Therefore, the potential of PSO to address high-dimensional FS problems still needs to be further explored.

Evolutionary multitasking [20], [21] is an emerging research topic in the EC. The main characteristic of evolutionary multitasking is that knowledge from different tasks are shared with each other to solve multiple optimization problems simultaneously. Evolutionary multitasking has been widely used to address different optimization problems in practice (i.e., travel salesman problem, job-shop scheduling problem, and parameter extraction problem) [22], [23]. This is because evolutionary multitasking has the promise of enhancing algorithm's search ability, accelerating convergence, and shortening computational time with the help of knowledge transfer between related tasks [24]–[26]. There are existing ways to generate related tasks for multitasking optimization in previous works, either by analyzing the characteristics of the fitness landscapes [27], [28] or simplifying the original investigated problems [24], [29]. However, the task generation methods based on the characteristics of the fitness landscape are not applicable for FS, since the information of fitness landscapes is not available. While the task generation methods based on the idea of simplifying the original problems are quite specific to their investigated problems (e.g., vehicle routing problem and signal reconstruction problem). In this article, we propose to construct multiple related tasks with different feature subsets to form a multitasking optimization problem. In other words, selecting features from different sets of the original features can be treated as a multitasking problem, since they share common features from the same dataset. For instance, if a dataset consists of ten features $\{F_1, F_2, \ldots, F_{10}\}$, the FS tasks that choose features from feature set $\{F_1, F_2, F_3, F_5, F_6, F_8, F_{10}\}$ or from feature set $\{F_1, F_3, F_4, F_5, F_7, F_{10}\}$, can be considered as related tasks in a multitasking scenario.

In our previous work [30], an evolutionary multitasking-based FS method named PSO-EMT was proposed on high-dimensional data. However, PSO-EMT can only handle two related tasks, and information sharing between the two tasks are achieved through a crossover operation. In addition, since one task contains all features, PSO-EMT still faces the problem of having high-computational cost. We will continue our previous work to investigate evolutionary multitasking further and develop a new FS method for solving high-dimensional classification problems. In comparison with previous FS approaches, the main contributions of this article can be described as follows.

1) Propose a new PSO-based multitasking approach for FS on high-dimensional data. The idea of solving a high-dimensional FS problem via evolutionary multitasking of several related low-dimensional FS problems significantly enhances the scalability of PSO on dealing with high-dimensional data.
2) Design a novel relevant task generation strategy based on the characteristics of the high-dimensional data. This strategy can adaptively generate several low-dimensional related FS tasks from a high-dimensional FS problem without losing much information about class labels.
3) Develop an effective knowledge transfer mechanism for sharing information among related tasks. In this strategy, knowledge from other related tasks is applied to help the current task jump out of local optima, thereby improving the quality of the current task solutions.

## II. Background and Related Work

### A. Feature Selection

Assuming that a labeled dataset contains $N$ samples and $M$ features (or attributes). Therefore, an FS problem can be expressed as: choosing $G$ features ($G<M$) from the original feature set to optimise the value of a given performance evaluation function $F(\cdot)$. For instance, in the classification problem, $F(\cdot)$ usually represents the classification error rate. During the FS process, a solution $X$ can be expressed as follows:

$$\begin{aligned} &X = (x_1, x_2, \ldots, x_M) \\ &x_m \in \{0, 1\} \quad \forall m \in \{1, 2, \ldots, M\} \end{aligned} \tag{1}$$

where $x_m = 1$ represents that the feature in the $m$th dimension should be selected; otherwise, the corresponding feature should be abandoned. Therefore, an FS problem with $M$ features can be described using the following formula:

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

$$\begin{aligned} &\min\ F(X) \\ &\quad \text{s.t.}\ \ X = (x_1, x_2, \ldots, x_M) \\ &\qquad x_m \in \{0, 1\} \quad \forall m \in \{1, 2, \ldots, M\}. \end{aligned} \tag{2}$$

### B. Evolutionary Multitasking Optimization

Evolutionary multitasking [31] is a relatively new research topic in the field of optimization and EC, and its propose is to improve global search performance through simultaneously optimizing multiple related tasks during the evolutionary process. In a multitasking optimization framework, solving one problem may facilitate tackling other problems. This is because some useful knowledge is complementary or shared between their problems. Mathematically, a multitasking optimization framework with $K$ tasks can be described as follows:

$$\begin{aligned} &\left\{x_1^*, x_2^*, \ldots, x_K^*\right\} \\ &\quad = \{\text{argmin} f_1(x_1), \text{argmin} f_2(x_2), \ldots, \text{argmin} f_K(x_K)\} \end{aligned} \tag{3}$$

where $x_k^*$ represents the optimal solution of Task $k$.

Multifactorial optimization (MFO) [20] is a novel paradigm in evolutionary multitasking, which can effectively search multiple decision spaces through implicit parallelism of population-based optimization algorithms to address the multiple related tasks simultaneously. In MFO, different tasks are encoded into a unified search space via a uniform random key scheme [32]. More details about MFO can be found in [20].

A number of evolutionary multitasking algorithms have been presented by the extending multifactorial evolutionary algorithm (MFEA) for solving different optimization problems [33]–[35]. For instance, an improved MFEA was developed for conducting evolutionary multitasking in [33], where a permutation-based unified representation and a split-based decoding operator were applied to the MFEA algorithm to address the NP-hard capacitated vehicle routing problems. In [34], an MFEA variant with multipopulation evolutionary framework was exploited for solving numeric optimization problems. A multifactorial genetic programming (MFGP) was proposed in [35], where a new scalable chromosome encoding scheme and a new evolutionary mechanism are designed for solving regression problems. However, most of them are still on the benchmark functions, only a few attempts for solving practical problems. In this study, we aim to develop a new multitasking optimization approach for solving the high-dimensional FS problems.

### C. Particle Swarm Optimization

In PSO [36], each particle denotes a candidate solution of the problem. In a $D$-dimensional search space, the $i$th particle at the $t$th iteration has two vectors to guide the search, which are the position vector $\boldsymbol{X}_i^t = [X_{i1}^t, X_{i2}^t, \ldots, X_{iD}^t]$ and the velocity vector $\boldsymbol{V}_i^t = [V_{i1}^t, V_{i2}^t, \ldots, V_{iD}^t]$, respectively. During the movement, the position and velocity of each particle are updated based on two positions (or solutions), one is the personal best position achieved by itself represented by $\boldsymbol{pbest}_i = [pbest_{i1}, pbest_{i2}, \ldots, pbest_{iD}]$, and the other is the global best position obtained by the entire population denoted by $\boldsymbol{gbest} = [gbest_1, gbest_2, \ldots, gbest_D]$. Based on *pbest* and *gbest*, the velocity and the position of the $i$th particle at the $(t+1)$th iteration are updated according to the following formulas:

$$\begin{aligned} V_{id}^{t+1} &= \omega * V_{id}^t + c_1 * r_1 * \left(p\text{best}_{id}^t - X_{id}^t\right) \\ &\quad + c_2 * r_2 * \left(g\text{best}_d^t - X_{id}^t\right) \end{aligned} \tag{4}$$

$$X_{id}^{t+1} = X_{id}^t + V_{id}^{t+1} \tag{5}$$

where $t$ denotes the current iteration number; $\omega$ represents the inertia weight; $c_1$ and $c_2$ denote acceleration constants called as the cognitive parameter and the social parameter, respectively; and $r_1$ and $r_2$ are uniform distributed values over the interval [0, 1].

### D. PSO-Based FS in Classification for High-Dimensional Data

In the last decades, researchers have developed a large number of PSO-based FS methods for the high-dimensional classification problems [16], [19], [37]. In these methods, PSO combines different search strategies to enhance the discriminative ability of the selected feature subset on high-dimensional classification data.

Reducing the search space before using PSO has become popular in FS on the high-dimensional classification problems. In [16], a variable-size cooperative coevolutionary PSO (VS-CCPSO) approach was proposed for FS. In this algorithm, a space division strategy was designed to divide a large feature space into multiple small subspaces according to the characteristics of data first. Then, a multiswarm PSO algorithm was used to search these feature subspaces cooperatively. Finally, a fitness-based elite combination strategy was applied to construct a complete feature subset from these subswarms. The results showed that this method is superior to other compared algorithms in terms of classification accuracy and the number of selected features. However, this method may fall into local optima due to ignoring interactions between different feature subspaces during the evolutionary process.

Designing a specific strategy based on the correlation information between features in PSO is an effective way to avoid invalid search during the FS process. In [19], a variable-length PSO algorithm was developed for FS on the high-dimensional classification problems. In this method, a variable-length representation strategy based on the relevance of features was presented to improve PSO's search capability in the different search space. The experimental results showed that this strategy can effectively improve the effectiveness of PSO during the search process. However, this strategy with different length representations might ignore some promising areas during the evolutionary process.

Utilizing a feature ranking method to eliminate irrelevant and redundant features is also proposed to enhance the scalability of PSO for FS in high-dimensional data. In [37], a hybrid model of filter and wrapper approaches was proposed for high-dimensional gene classification problems. In this approach, a correlation-based filter method was used to determine an initial feature subset, and then PSO was applied to choose

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

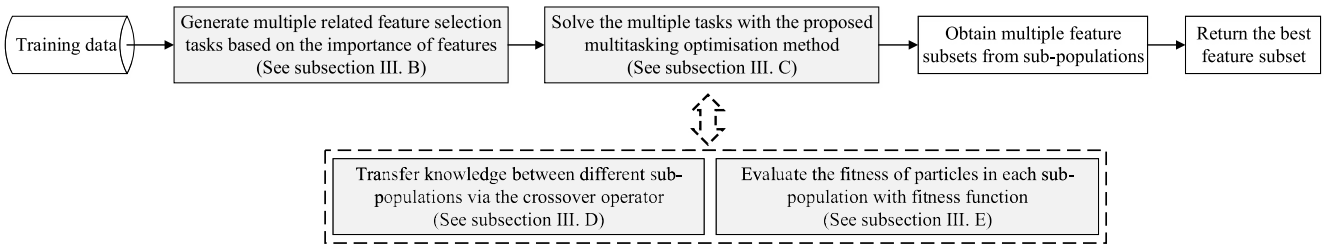



Fig. 1. Basic framework of the proposed algorithm.

features from the initial feature subset. The experimental results showed that this method is superior to other compared approaches in terms of classification accuracy and computational time. However, based solely on the correlation of each feature with the class labels is not enough to determine an optimal initial subset since they ignore feature interactions.

Developing local search strategies to exploit the promising areas is an effective way to promote the search capability of PSO on high-dimensional problems. In [38], PSO with a local search mechanism on *pbest* was proposed for high-dimensional FS problems. In this method, the local search mechanism was used as a complimentary search mechanism of the PSO-based global search, which can exploit the space near the individual best positions. The results showed that this strategy can significantly enhance the classification accuracy on the high-dimensional FS problems in most cases. However, this FS method usually requires more computing resources to enhance the local search of the particles, especially when the search space is large.

In summary, although many PSO-based methods have been proposed and have shown promise for high-dimensional classification problems, FS in high-dimensional data is still an open challenge due to the high-computational cost and easy to fall into local optima. Multitasking optimization have shown promise on many challenging problems. In this study, we investigate how multitasking optimization can be used to improve FS performance on high-dimensional data.

## III. Proposed Approach

### A. Framework of the Proposed Approach

Evolutionary multitasking is an effective technique to deal with complex problems through performing knowledge transfer between multiple related tasks. Inspired by this idea, we design a new FS method named MTPSO by utilizing the advantages of multitasking optimization approach to address high-dimensional FS classification as a multitasking problem.

The main framework of MTPSO is shown in Fig. 1. The input is the training data including all available features. The output is the feature subset obtained by the subpopulations on multiple related tasks. MTPSO consists of two parts, that is, generating multiple related FS tasks according to the characteristics of the high-dimensional data and solving these multiple related FS tasks via knowledge transfer. When receiving a high-dimensional data, a feature ranking method is first adopted to measure the importance of features, and then a method based on the importance of features is performed to produce multiple related FS tasks. Next, each FS task is assigned a subpopulation, and the proposed multitasking optimization method is employed to search these related FS tasks via knowledge transfer, until the stopping criterion is met. In the end, a feature subset with the highest classification accuracy from the multiple feature subsets is selected as the final feature subset. If more than one feature subsets achieve the same highest classification accuracy, the one with the smallest number of selected features will be chosen as the final feature subset.

Four research questions are involved in this study. The first is how to develop a strategy to generate related FS tasks without losing too much information about the class labels. The second is how to design a method to effectively deal with multiple related FS tasks through multitasking optimization. The third is how to design an effective knowledge transfer mechanism in the proposed multitasking optimization method to implement positive information sharing between different tasks. The last is how to determine an effective fitness function to accurately evaluate the selected feature subsets. The details of the proposed method to address these four research questions are described in the next sections.

### B. Task Generation Strategy

In FS tasks, the useful information is mainly the relevance of features and the interactions between features. In the proposed method, the tasks are generated based on the relevance of features, i.e., highly relevant features have a high probability of being chosen to form a task (as the candidate feature set). However, due to feature interactions, the optimal feature subset is usually a set of complementary features, including both highly relevant and weakly relevant features. Therefore, weakly relevant features also have a chance (low probability) to be chosen to preserve feature interaction information in a task. Although not all information of the original task is preserved, the most important information is very likely to be preserved in different tasks. Using evolutionary multitasking is on those generated tasks are expected to improve the performance. The detailed description of the generation task strategy is described as follows.

First, we use *ReliefF* [39] to evaluate the relevance information of features. This is because *ReliefF* can better determine the nonlinear relationship between variables than other correlation measures, and it has been widely adopted to determine the importance of a feature to the class labels by

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

**Algorithm 1:** Knee Point Selection Strategy

**Input**: The all available features.
**Output**: Two feature sets *ProFeatures* and *RemFeatures*.
1 Calculate feature weights $W$ using *ReliefF*.
2 Sort $W$ based on descending order.
3 *maxPoint*(1,$W$(1)).
4 *minPoint*(*numFeature*,$W$(*numFeature*)).
5 Set *maxDistance* $\leftarrow 0$ and *kneePoint* $\leftarrow 0$.
6 Get a line $L$ based on *maxPoint* and *minPoint* points.
7 **for** $i$ = *1 to numFeature* **do**
8  Calculate distance $d$ from *Point*$(i, W(i))$ to line $L$.
9  **if** $d >$ *maxDistance* **then**
10  *maxDistance* $\leftarrow d$.
11  *kneePoint* $\leftarrow i$.
12  **end**
13 **end**
14 **for** $i$ = *1 to numFeature* **do**
15  **if** $i <=$ *kneePoint* **then**
16  *ProFeatures* $\leftarrow$ *Feature*$(i)$.
17  **end**
18  **else**
19  *RemFeatures* $\leftarrow$ *Feature*$(i)$.
20  **end**
21 **end**
22 return *ProFeatures*, *RemFeatures*.

assigning a weight to each feature. In *ReliefF*, the weights are determined based on the distinguishing ability of features to the close-range samples. When using ReliefF to handle multi-class problems, one sample $R$ is first randomly selected from the training dataset, and then it finds $h$ nearest neighbors of the sample $R$ from the same and different classes, which are denoted by $H_j$ $(j = 1, 2, \ldots, h)$ and $M_j(c)$ $(j = 1, 2, \ldots, h)$, respectively. Equation (6) is used to calculate the weight of feature $a$. The larger the weight, the more important the feature

$$W(a)^{t+1} = W(a)^t - \sum_{j=1}^{h} f\left(a, R, H_j\right)/(u * h) + \sum_{c \notin s(R)} \left[ \frac{p(c)}{1 - p(s(R))} * \sum_{j=1}^{h} f\left(a, R, M_j(c)\right) \right] \Big/ (u * h) \quad (6)$$

where $W(a)$ is the weight of feature $a$; $t$ denotes the number of current iteration; $u$ represents the number of sampling times; $s(R)$ is the class label of the sample $R$; and $p(c)$ and $p(s(R))$ represent the proportion of class $c$ and the class of $R$, respectively. $f(a, R_1, R_2)$ means the difference between the values of the feature $a$ for samples $R_1$ and $R_2$.

Second, we propose a knee point selection strategy based on the importance of features for dividing all available features into two feature sets (i.e., the promising feature set and the remaining feature set). The use of the largest distance to the extreme line to characterize knee point was first proposed by Das [40], which has been widely applied to determine a demarcation point in many optimization problems [41]. Inspired by this idea, we propose a knee point selection strategy based on the weights of features, which can reduce one parameter (i.e., the threshold value for dividing features) in the proposed method. The pseudo code of dividing all features into two sets is shown in Algorithm 1. In the knee point selection strategy, features are sorted in a descending order based on their weights

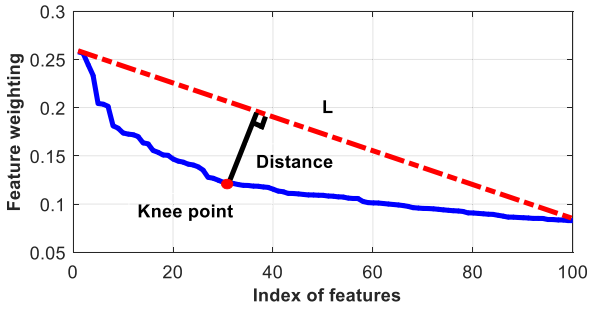



Fig. 2. Example is used to illustrate the determination of the knee point.

calculated by ReliefF (from lines 1 and 2), then a curve about the weights is achieved. Next, a line ($L$) is generated between two point with maximal and minimal feature weights (from lines 3–6). The distances between each point on the curve and $L$ are calculated, and the point that has the largest distance to $L$ is detected as the knee point (from line 7–13). Fig. 2 is an example to illustrate the determination of the knee point, where the dataset has 100 features. Finally, the features whose weights are larger than that of the knee point are selected to form the promising feature set (i.e., *ProFeatures*), and the remaining features are assigned to the remaining feature set (i.e., *RemFeatures*) (from lines 14–21).

Third, a task generation strategy based on the promising feature set and the remaining feature set is developed to form related FS tasks. Intuitively, the promising feature set is more beneficial for FS than the remaining feature set. Since ReliefF cannot effectively remove redundant features, the promising feature set still contains redundant features. Furthermore, some features in the remaining feature set are useful due to the interactions between features. Therefore, a probability is used to select features from the promising feature set and the remaining feature set to form related FS tasks. In the proposed strategy, the average feature weights of the promising feature set and the remaining feature set are used to determine the selection probability. The formula for calculating the selection probability is described as follows:

$$p = \frac{W_p}{W_p + W_r} \quad (7)$$

where $W_p$ and $W_r$ represent the average weights of features in the promising feature set and the remaining feature set, respectively.

Suppose a dataset has $n$ features, where $m$ features belong to the promising feature set. Fig. 3 shows the process of using the proposed task generation strategy to generate an FS task. If features are in the promising feature set, when condition *rand*$<= p$ holds, the corresponding feature is selected into *Task*, otherwise it is unselected. If features are in the remaining feature set, when condition *rand*$< (1-p)$ is activated, the corresponding feature is selected into *Task*, otherwise it is unselected, where *rand* indicates a random value between [0, 1]. $T$ different FS tasks can be obtained by repeating the above process $T$ times. Since the selection probability is determined by taking the relevance of features into account, the individually useful features are more likely to be preserved. In particular,

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

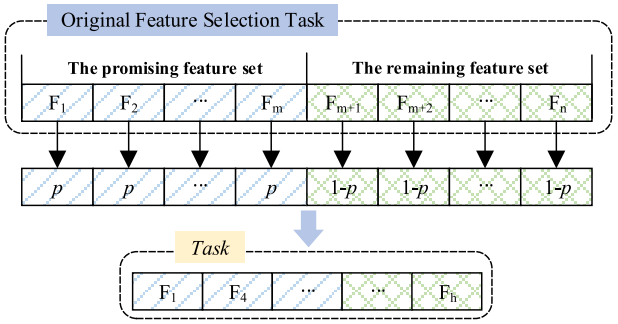



Fig. 3. Process of generating tasks through the proposed task generation strategy.

the features in the promising feature set have a high probability of being chosen in the generated FS task, and the features in the remaining feature set still have a chance to be chosen. The proposed strategy is expected to avoid wasting computational resources on exploring undesired regions (e.g., the regions with irrelevant features) by transforming the huge search space to multiple small but promising spaces (i.e., multiple related tasks).

Although it is not guaranteed that the optimal feature subsets are included in these generated related FS tasks, it is quite likely that an optimal feature subset can still be found by the proposed method due to the following reasons. First, FS problems often have more than one optimal solutions due to the existence of redundant features and feature interactions. Second, in the proposed task generation strategy, both highly and weakly relevant features have a chance to be included in the generated FS tasks, where highly relevant features that are more likely to form the optimal feature subset, are more likely to be included. Third, there are overlaps between the candidate features in these related tasks, so that important interactions between useful features can be mostly kept and discovered. Last but not the least, these related FS tasks are not solved independently, and useful information is shared between related tasks to utilize the multitasking idea to improve the FS performance.

### C. Multitasking Optimization Method

Multipopulation is an effective technique to divide the whole population into multiple subpopulations to search multiple different areas simultaneously. Multipopulation combined with different EC algorithms have been presented to solve multitasking optimization [42], [43]. Based on multipopulation technique, we develop a new multitasking optimization method for high-dimensional FS problems. In the proposed method, each subpopulation corresponds to a task. Unlike other multipopulation multitasking optimization methods, a new knowledge transfer mechanism is designed to realize the communication between different tasks (subpopulations). The details of the designed knowledge transfer mechanism will be introduced in the next section. Since PSO has strong global search ability and a natural representation for FS, the multitasking optimization method based on PSO is developed.

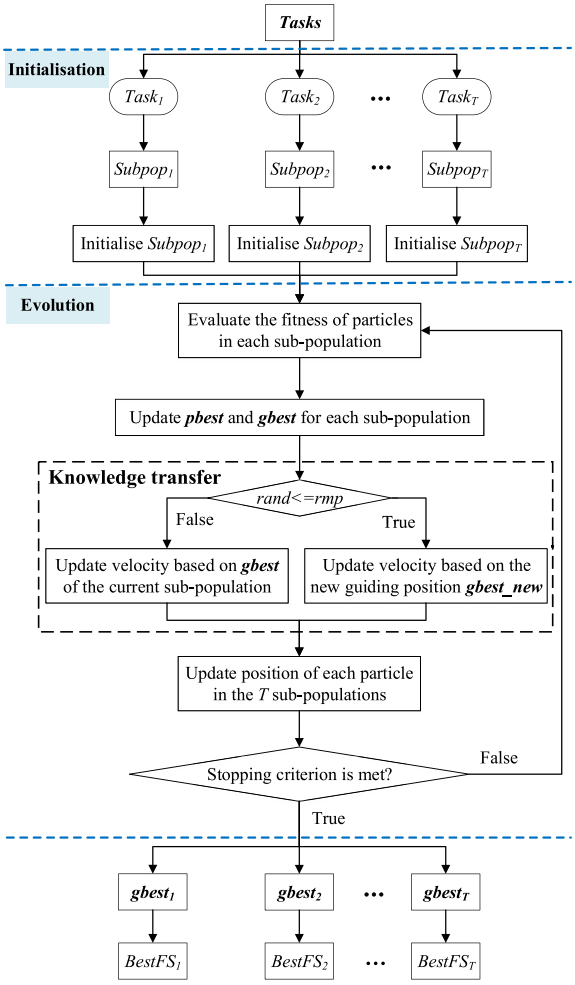



Fig. 4. Evolutionary process of the proposed multitasking method.

Fig. 4 shows the evolutionary process of the proposed multitasking optimization method, which consists of two stages (i.e., *initialization* and *evolution*). Supposing $T$ related FS tasks ($Task_1$, $Task_2$, ..., $Task_T$) with different feature spaces are generated to find an optimal feature subset for the desired high-dimensional classification problem via knowledge transfer between these related FS tasks. In the initialization, $T$ subpopulations (e.g., $Subpop_1$, $Subpop_2$, ..., $Subpop_T$) are generated for FS tasks, and each subpopulation aims to address its corresponding FS task. During the evolutionary process, FS tasks located in different subpopulations will help each other through knowledge transfer. Therefore, the proposed multitasking optimization method will output $T$ feature subsets (e.g., $BestFS_1$, $BestFS_2$, ..., $BestFS_T$) from $T$ subpopulations.

There are several potential benefits of using the developed multipopulation multitasking optimization method to implement FS on the high-dimensional classification problems. First, the developed multitasking framework makes it possible to solve the high-dimensional classification through knowledge transfer between several related FS tasks. Second, knowledge from other FS tasks with different feature spaces is used to help the subpopulation to jump out of possible local optima, enabling the current FS task to find a better feature subset.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

Third, this way of dealing with the high-dimensional classification can greatly reduce the computational time since the search spaces of these subpopulations are much smaller than the original search space.

### D. Knowledge Transfer Mechanism

PSO is an effective population-based optimization algorithm, which is usually used to solve a single task. Multitasking is an effective approach for solving multiple tasks simultaneously. Therefore, when extending PSO for multitasking, two key issues need to be considered. The first is how to determine what knowledge that can be transferred between different tasks. The second is how to achieve effective knowledge transfer between different tasks. To solve the above issues, this study proposes a new knowledge transfer mechanism for information sharing between related tasks. The details are shown as follows.

*How to Transfer:* The way of transferring knowledge to other tasks has a great impact on the quality of the final solution. If a poor transfer operator is adopted to implement information sharing between related tasks, it will reduce the purpose of using knowledge transfer to enhance search quality during the evolutionary process. Crossover is an important operator, which has been widely used for multitasking optimization to realize information sharing and its effectiveness has been demonstrated [26], [34]. In addition, using the crossover to implement knowledge transfer will not add much extra cost. In this study, the crossover operator on the particle's position is used as the way of knowledge transfer between different tasks (i.e., subpopulations). Sharing information by crossover can not only implement knowledge transfer between different tasks, but also enhance diversity during the search process.

*When to Transfer:* During the evolutionary process, we define a random mating probability (rmp) to control when to transfer knowledge from other tasks. A large rmp encourages knowledge transfer from other tasks, and vice versa. At each generation, if the knowledge transfer strategy is activated ($rand <= rmp$), knowledge from other tasks will be used to assist in updating the positions of particles. Otherwise ($rand > rmp$), the positions of particles are updated by knowledge from the current task.

*What to Transfer:* It is critical to determine what kinds of knowledge are beneficial to be shared. In PSO, knowledge carried by the global optimal position plays an important role in the velocity updating, which is used to influence the search direction of each particle during the search process. Therefore, we use crossover on the global optimal position to implement knowledge transfer between different tasks. To be specific, for each particle in $Task_t$, the tournament selection mechanism [44] is used to select a *gbest* from all *gbest* of other subpopulations, and the selected *gbest* is recorded as $gbest^m$. A new guiding position based on these two global optimal solutions is generated through arithmetic crossover [45]. For $Task_t$, the process of generating the new guiding position is shown as follows:

$$g\text{best_new}^t = cr * g\text{best}^t + (1 - cr) * g\text{best}^m \tag{8}$$

where $gbest^t$ represent the current global optimal positions of $Task_t$. $cr$ denotes a random vector, and the range of each dimension is between 0 and 1.

During the evolutionary process, if $gbest^t$ does not change within the given generations ($G$), PSO in $Task_t$ may be trap into a potential local optimum. In this study, we propose a new operator to generate the new guiding position ($gbest_new^t$) based on the average of $T$ global optimal positions. This operator is expected to find a new search region, thereby increasing the possibility of escaping from the local optima during the search process. Equation (9) shows the formula of this operator for $Task_t$

$$gbest_new^t = \frac{1}{T} * \sum_{i=1}^{T} gbest^i \tag{9}$$

where $gbest^i$ represents global optimal position so far in $Task_i$. $T$ denotes the number of tasks.

In this way, particles in different tasks have a chance to share knowledge with the other by *gbest*. During the evolutionary process, if a random event ($rand <= rmp$) is activated, *gbest_new* will replace the original *gbest* to guide the velocity updating.

### E. Fitness Function

Feature evaluation is a critical component that assesses the quality of the selected feature subset to guide the evolutionary process. There are two main objectives in FS, which are to maximize the classification accuracy and to minimize the feature subset size. In this study, a fitness function combined by these two objectives is applied to evaluate the selected features. Equation (10) shows the fitness function, which uses a weight to combine the classification error rate and the number of selected features

$$\text{fitness} = \alpha * \text{ErrorRate} + (1 - \alpha) * \frac{\#\text{Selected}}{\#\text{All}} \tag{10}$$

where ErrorRate represents the classification error rate of the learning algorithm, #Selected denotes the number of selected features, and #All indicates the total number of available features in the training dataset. $\alpha$ is the weight to control the contribution of the classification error rate and the number of selected features. According to [46], the range of $\alpha$ is [0, 1]. Since the classification performance is preferred than the feature subset size, we set $\alpha = 0.9$ in this article.

Since most of the collected data are unbalanced, a balance accuracy [47] is applied to evaluate the first component of the fitness function in this study. 5-fold cross-validation on the training samples is applied to avoid FS bias during the training process. The calculation of the classification error rate is described as follows:

$$\text{ErrorRate} = 1 - \frac{1}{c} * \sum_{i=1}^{c} \text{TPR}_i \tag{11}$$

where $c$ denotes the number of classes in the data, and $\text{TPR}_i$ represents the true positive rate, i.e., the proportion of correctly identified instances in class $i$. Since the balanced accuracy has no bias to each class in the classification problem, the weight for each class is set to $1/c$.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

TABLE I
BASIC INFORMATION OF THE 18 DATASETS

| Datasets | #Features | #Ins. | #Class | %Smallest class | %Largest class |
|---|---|---|---|---|---|
| SRBCT | 2,308 | 83 | 4 | 13 | 35 |
| PCMAC | 3,289 | 1,943 | 2 | 49 | 51 |
| Lung Cancer 1 | 3,312 | 203 | 5 | 3 | 68 |
| RELATHE | 4,322 | 1,427 | 2 | 45 | 55 |
| BASEHOCK | 4,862 | 1,993 | 2 | 50 | 50 |
| Leukemia 1 | 5,327 | 72 | 3 | 13 | 53 |
| DLBCL | 5,469 | 77 | 2 | 25 | 75 |
| 9Tumor | 5,726 | 60 | 9 | 3 | 15 |
| Brain Tumor 1 | 5,920 | 90 | 5 | 4 | 67 |
| Prostate Tumor 1 | 5,966 | 102 | 2 | 49 | 51 |
| Leukemia 2 | 7,129 | 72 | 4 | 6 | 53 |
| Brain Tumor 2 | 10,367 | 50 | 4 | 14 | 30 |
| Prostate Tumor 2 | 10,509 | 102 | 2 | 49 | 51 |
| Leukemia 3 | 11,225 | 72 | 3 | 28 | 39 |
| 11Tumor | 12,533 | 174 | 11 | 4 | 16 |
| Lung Cancer 2 | 12,600 | 203 | 5 | 3 | 68 |
| SMK-CAN-187 | 19,993 | 187 | 2 | 48 | 52 |
| GLI-85 | 22,283 | 85 | 2 | 31 | 69 |

## IV. EXPERIMENT DESIGN

To investigate the effectiveness (i.e., the classification accuracy and the number of selected features) and efficiency (i.e., the training time) of the proposed FS method, a set of experiments have been conducted. This section describes the investigated datasets, the comparative methods, the parameter settings, and the sensitivity analysis of key parameters.

### A. Datasets

To evaluate the performance of the proposed FS method, 18 datasets with thousands of features are selected in the experimental analysis. They are open sources on http://featureselection.asu.edu. Table I shows their basic information. In addition, these datasets are highly imbalanced. Therefore, classification on these datasets is a difficult task.

### B. Comparative Methods and Parameter Settings

To evaluate the proposed approach, the following state-of-the-art FS approaches are selected as compared methods.

1) Chaotic binary PSO FS (CBPSO) [48].
2) PSO-based FS with novel initialization and updating strategies [PSO(4-2)] [6].
3) Hybrid PSO FS based on the spiral-shape mechanism (HPSO-SSM) [46].
4) Sticky binary PSO-based FS with the dynamic strategy (SBPSO-D) [49].
5) PSO-based FS with the variable-length strategy (VLPSO) [19].
6) VS-CCPSO approach for FS [16].
7) MFEA [20].
8) Differential evolution (DE) [50].
9) The genetic algorithm (GA) [51].

The maximum number of iterations (*max_iter*) is set to 100 for the six approaches. Since the datasets in Table I have a different number of features ranging from 2308 to 22 283, we set the size of the population to 1/30 of the number of features for all algorithms to maintain PSO's efficiency in the large search space, and we limit it to 200 to avoid the high-computational cost. Table II shows the parameter settings of these ten comparison algorithms, where *iter* denotes the current iteration. It is noted that the parameter setting of these compared algorithms are according to its corresponding reference. During the FS process, 10-fold cross-validation is adopted to divide training and test set. That is, the original dataset is divided into ten groups. One group is used as the test samples and the rest nine groups are employed as the training samples, this process is repeated ten times. To avoid FS bias, an inner loop of 5-fold cross-validation is applied on the training set to evaluate the classification performance of selected features during the evolutionary process. It is noted that only the training samples are used in this process. After the training process, the selected features are applied to assess the classification accuracy of the proposed FS method on the unseen test samples.

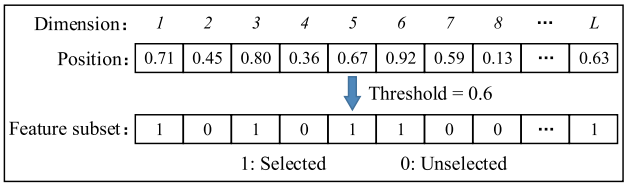



Fig. 5. Representation of an MTPSO particle with a dimension $L$.

Since the new position element in binary PSO is determined without considering its previous position, continuous PSO can usually achieve better performance than binary PSO for FS problems as shown in [49]. In this study, the proposed MTPSO method is based on continuous PSO. The representation of each particle in MTPSO is shown in Fig. 5. It is a $L$-dimensional vector, where $L$ represents the number of features in the dataset. In MTPSO, a particle's position represents a potential feature subset, and the range of each dimension is within [0, 1]. During the search process, a threshold $\theta$ is used to determine which feature is selected, that is, if the value in the vector is larger than $\theta$, the corresponding feature is selected. Otherwise, it is not selected. Following the suggestion in [52], the threshold is set to 0.6 in this study. In the proposed method, the $k$-nearest neighbor (KNN) [53] algorithm is used as a classifier due to its simplicity and effectiveness, where $K$ is set to 5 [54].

### C. Sensitivity Analysis of Parameters

In the proposed method, there are three important parameters: 1) rmp is an important parameter in the knowledge transfer mechanism, which is used to control when to transfer; 2) the predetermined generation ($G$) is used to change the velocity updating in the knowledge transfer mechanism; and 3) the number of tasks ($T$) is an important parameter of the proposed method. In this section, we study the effect of these three parameters.

For rmp and $G$, the experimental analysis are conducted on the Brain2 dataset because it has a medium number of features compared with other examined problems. In this experiment,

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PARAMETER SETTING

| Algorithms | Parameter values |
|---|---|
| CBPSO | The acceleration coefficients $c_1$ = $c_2$ = 2, the inertia weight $w(t+1)$ = 4*$w(t)$*(1-$w(t)$), where $w(0)$ = 0.48. |
| PSO(4-2) | $c_1$ = $c_2$ = 1.49618, $w$ = 0.7298. |
| HPSO-SSM | $c_1$ = $c_2$ = 2, $w$ = 0.9-0.5*(iter/max_iter), the selection probability $p$ = 0.8. |
| SBPSO-D | The fix number of steps $ustkS^L$ = 1*max_iter/100, and $ustkS^U$ = 10*max_iter/100, the stickiness factors $i_s^L$ = 0/n, and $i_s^U$ = 10/n, the ratio $\alpha$ = 2. |
| VLPSO | $c$ = 1.49445, $w$ = 0.9-0.5*(iter/max_iter), the max iterations for renew exemplars $\alpha$ = 7, the number of divisions $\eta$ = 12, the max iterations for length changing $\theta$ = 9. |
| VS-CCPSO | The probability to control convergence speed $prob$ = 0.7, the number of feature subspaces $M$ = 8. |
| MFEA | The probability of individual learning $pil$ = 1, random mating probability $rmp$ = 0.6, the standard deviation of gaussian mutation model $gm$ = 0.02, the selection pressure is $roulette\ wheel$. |
| DE | The mutation rate $m$ = 0.6, the crossover rate $cr$ = 0.9. |
| GA | The probability of mutation $pm$ = 0.001, the probability of crossover $pc$ = 1. |
| MTPSO | $c_1$ = $c_2$ = 1.49445, $w$ = 0.9-0.5*(iter/max_iter), random mating probability $rmp$ = 0.6, the max iterations for updating velocity using the united information $G$ = 6. |

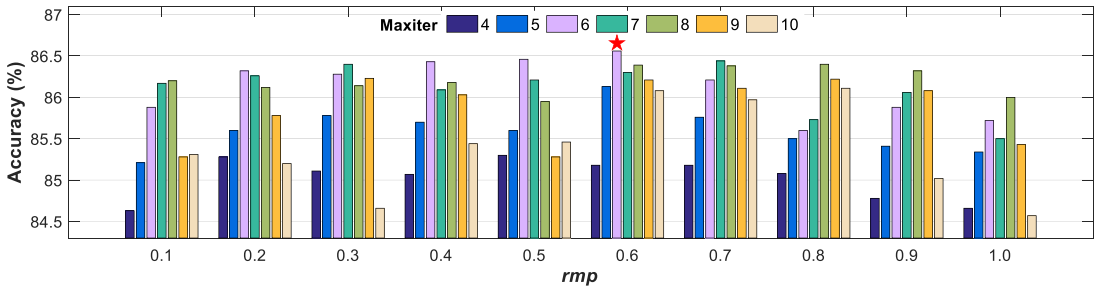



Fig. 6. Average classification accuracies of 70 combinations of two parameters (rmp and maximum iterations to update velocity using united information).

MTPSO is conducted ten different values for rmp ranging from 0.1 to 1, and seven vales from 4 to 10 for *G*, generating 70 combinations of these two parameters. Each combination is conducted 30 times independently. Fig. 6 shows the average classification accuracy for each combination, and an five-pointed star ($\star$) is used to point the best result. Fig. 6 shows that the best combination is {*rmp* of 0.6, *G* of 6}. In this study, the {*rmp* of 0.6, *G* of 6} is used in MTPSO for all datasets. For *T*, we determine it based on an experimental analysis. Due page limitations, we put the experimental details into the supplementary materials.

## V. RESULTS AND DISCUSSIONS

In this section, several experiments are conducted to illustrate the effectiveness and efficiency of the proposed approach. The Wilcoxon rank-sum test with a significance level of 0.05 is adopted to evaluate the performance of the proposed FS method with 30 independent runs. In the following results, "+," "–," and "≈" indicate that the proposed FS method is significantly better than, worse than, and similar to the compared methods. The larger the number of "+" symbols, the better the proposed method is.

### A. Classification Performance of MTPSO

The classification accuracy and the number of selected features are important indicators to measure the effectiveness of the FS approach. In this experiment, high-classification accuracy, and a small number of features indicate that the algorithm is superior for solving FS problems.

Table III shows the average (AvgAcc) and standard deviation (Std) of the classification accuracies of MTPSO and the other nine compared algorithms based on the 30 independent runs on the 18 high-dimensional classification datasets. The bold numbers mean that the corresponding method obtains the best result. As can be seen from Table III, the average classification accuracies of the proposed MTPSO method are better than those of the compared methods on 16 out of the 18 datasets. For example, the proposed method achieves 100% classification accuracy in all the 30 independent runs on the SRBCT dataset, which is 10.49% more than that of the DE algorithm. Compared with the VLPSO algorithm, MTPSO achieves the highest improvement on the Brain2 dataset, where the average classification accuracy is increased by 20.49%. In terms of the significance tests, the proposed MTPSO method wins 157, draws 5, and loses 0 in the 162 comparisons. This confirms the effectiveness of the proposed multitasking-based FS method.

Table IV compares MTPSO with the other nine algorithms in terms of the number of selected features. As can be seen from Table IV, VLPSO selects fewer features than that of MTPSO on 15 out of the 18 datasets. The main reason is that VLPSO uses a length-changing mechanism to reduce the number of selected features during the evolutionary process. However, the fast removal of features may affect the classification performance of VLPSO. The classification accuracy in Table III also confirms that the average classification accuracies achieved by MTPSO are much higher than VLPSO on 17 out of the 18 datasets. In comparisons with the remaining eight methods (i.e., CBPSO, PSO(4-2),

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

TABLE III
AVERAGE AND STD OF THE CLASSIFICATION ACCURACY (AVGACC±STD) OBTAINED BY THE TEN ALGORITHMS OVER THE 30 INDEPENDENT RUNS ON THE 18 TEST SETS

| Datasets | CBPSO | PSO(4-2) | HPSO-SSM | SBPSO-D | VLPSO | VS-CCPSO | MFEA | DE | GA | MTPSO |
|---|---|---|---|---|---|---|---|---|---|---|
| SRBCT | 93.53±2.59(+) | 95.37±2.24(+) | 94.76±2.82(+) | 93.51±1.64(+) | 99.67±0.52(≈) | 97.43±1.57(+) | 98.30±0.82(+) | 89.51±4.47(+) | 90.86±4.32(+) | **100.00±0.00** |
| PCMAC | 77.48±1.74(+) | 77.44±1.20(+) | 77.12±1.65(+) | 76.97±1.75(+) | 82.12±0.80(+) | 81.94±0.83(+) | 80.99±1.17(+) | 77.05±1.08(+) | 78.39±2.26(+) | **87.84±0.21** |
| Lung1 | 95.21±0.82(+) | 94.72±0.92(+) | 94.60±1.05(+) | 94.86±1.07(+) | 90.77±1.41(+) | **97.91±0.28**(≈) | 95.24±1.56(+) | 95.19±1.59(+) | 96.08±1.60(+) | 97.40±0.54 |
| RELATHE | 81.31±1.65(+) | 79.43±1.82(+) | 81.24±1.79(+) | 81.60±2.36(+) | 71.77±0.81(+) | 82.20±0.55(≈) | 73.89±2.16(+) | 76.54±2.89(+) | 75.22±0.49(+) | **83.07±0.16** |
| BASEHOCK | 85.97±1.83(+) | 84.20±2.17(+) | 85.02±2.05(+) | 85.43±1.98(+) | **94.13±1.98**(≈) | 90.91±0.71(+) | 87.98±1.54(+) | 84.57±1.26(+) | 86.58±1.75(+) | 93.29±0.28 |
| Leuk1 | 89.52±1.91(+) | 89.35±2.85(+) | 89.88±2.78(+) | 88.91±2.56(+) | 93.31±2.34(+) | 96.27±0.25(+) | 95.65±2.37(+) | 91.45±3.12(+) | 92.90±4.19(+) | **98.24±0.81** |
| DLBCL | 88.78±2.30(+) | 89.57±2.35(+) | 89.41±3.86(+) | 89.35±2.10(+) | 86.51±2.88(+) | 95.76±1.60(+) | 97.91±1.03(≈) | 88.89±5.62(+) | 88.47±6.06(+) | **98.89±0.81** |
| 9Tumor | 45.78±4.73(+) | 47.33±3.17(+) | 45.94±5.06(+) | 45.39±3.14(+) | 54.94±4.80(+) | 64.11±3.15(+) | 65.45±2.61(+) | 44.85±7.14(+) | 47.88±6.62(+) | **69.06±2.34** |
| Brain1 | 85.26±1.27(+) | 84.33±2.17(+) | 84.37±2.10(+) | 85.15±1.32(+) | 71.19±3.52(+) | 87.15±0.38(+) | 85.71±1.73(+) | 85.36±3.67(+) | 85.83±2.73(+) | **89.48±1.56** |
| Prostate1 | 85.46±2.20(+) | 85.91±2.03(+) | 85.64±2.53(+) | 85.70±1.79(+) | 88.74±2.23(+) | 90.41±0.78(+) | 89.77±1.09(+) | 76.24±4.83(+) | 77.63±4.93(+) | **92.77±0.68** |
| Leuk2 | 84.54±2.16(+) | 82.81±2.96(+) | 84.97±3.20(+) | 84.71±2.67(+) | 85.82±2.96(+) | 88.10±1.73(+) | 87.50±2.74(+) | 77.50±3.39(+) | 80.42±2.71(+) | **92.05±0.97** |
| Brain2 | 71.33±4.01(+) | 71.00±3.81(+) | 71.60±3.94(+) | 71.87±3.89(+) | 66.78±4.10(+) | 80.47±2.39(+) | 77.78±3.68(+) | 71.30±5.48(+) | 70.01±5.18(+) | **85.40±2.47** |
| Prostate2 | 81.47±2.02(+) | 84.98±2.33(+) | 84.23±2.44(+) | 81.35±2.46(+) | 89.82±2.28(+) | 89.63±0.92(+) | 88.32±2.53(+) | 77.53±3.74(+) | 78.92±4.04(+) | **91.77±0.96** |
| Leuk3 | 91.99±2.25(+) | 92.48±2.32(+) | 92.20±2.77(+) | 91.82±2.00(+) | 91.56±1.67(+) | 94.77±1.53(+) | 93.51±1.42(+) | 95.23±1.89(+) | 94.09±2.09(+) | **97.43±1.12** |
| 11Tumor | 83.92±1.39(+) | 82.48±1.97(+) | 82.66±1.43(+) | 83.40±1.47(+) | 80.92±2.39(+) | 88.01±0.69(+) | 81.03±2.93(+) | 79.14±3.66(+) | 80.92±4.20(+) | **91.86±1.03** |
| Lung2 | 92.99±1.39(+) | 92.41±1.31(+) | 93.17±1.31(+) | 92.75±0.89(+) | 89.55±1.68(+) | 92.96±0.80(+) | 93.65±1.24(+) | 92.17±2.08(+) | 92.65±1.59(+) | **96.67±0.52** |
| SMK187 | 63.83±1.53(+) | 65.05±2.09(+) | 64.05±1.93(+) | 64.13±1.37(+) | 63.48±2.69(+) | 71.08±0.95(+) | 70.69±2.03(+) | 63.91±2.17(+) | 65.34±2.65(+) | **75.96±1.01** |
| GLI-85 | 81.81±2.42(+) | 82.79±2.72(+) | 82.66±2.72(+) | 81.45±2.27(+) | 81.42±2.26(+) | 87.61±1.37(+) | 88.46±2.39(+) | 79.74±4.03(+) | 80.13±2.87(+) | **92.77±1.38** |

TABLE IV
NUMBER OF FEATURES SELECTED BY THE TEN ALGORITHMS ON THE 18 DATASETS

| Datasets | CBPSO | PSO(4-2) | HPSO-SSM | SBPSO-D | VLPSO | VS-CCPSO | MFEA | DE | GA | MTPSO |
|---|---|---|---|---|---|---|---|---|---|---|
| SRBCT | 979.93 | 295.25 | 273.30 | 1128.13 | **49.08** | 273.83 | 152.03 | 898.73 | 889.60 | 126.50 |
| PCMAC | 1522.37 | 476.03 | 976.70 | 1638.60 | **57.93** | 874.00 | 67.13 | 1302.60 | 1290.40 | 62.47 |
| Lung1 | 1446.73 | 472.64 | 643.89 | 1640.61 | 370.79 | 566.60 | 357.07 | 1299.37 | 1303.03 | **343.24** |
| RELATHE | 2038.57 | 757.30 | 1406.67 | 2147.10 | 133.43 | 950.67 | 102.63 | 1722.60 | 1692.40 | **83.60** |
| BASEHOCK | 2369.13 | 1302.17 | 1744.70 | 2425.93 | 180.77 | 1044.90 | 132.83 | 1927.00 | 1924.80 | **118.83** |
| Leuk1 | 2413.93 | 762.12 | 940.04 | 2643.68 | **54.70** | 846.03 | 352.73 | 2102.87 | 2037.23 | 317.00 |
| DLBCL | 2327.41 | 739.03 | 658.58 | 2691.41 | **48.14** | 662.60 | 280.56 | 2147.07 | 2083.93 | 237.36 |
| 9Tumor | 2676.92 | 898.03 | 1822.62 | 2858.77 | **47.05** | 747.07 | 435.33 | 2285.93 | 2273.70 | 423.44 |
| Brain1 | 2599.66 | 863.54 | 1102.05 | 2933.21 | **26.83** | 1453.17 | 593.83 | 2347.23 | 2286.90 | 587.00 |
| Prostate1 | 2554.21 | 821.66 | 883.07 | 2950.99 | **35.97** | 798.77 | 435.47 | 2360.83 | 2337.33 | 368.09 |
| Leuk2 | 3270.94 | 1067.70 | 1681.10 | 3548.16 | **53.39** | 1207.70 | 437.09 | 2823.83 | 2738.17 | 434.84 |
| Brain2 | 4574.74 | 1537.84 | 2551.54 | 3769.82 | **81.46** | 1752.10 | 1069.18 | 4115.80 | 4119.50 | 1066.32 |
| Prostate2 | 4679.56 | 1472.59 | 1331.75 | 4227.83 | **26.44** | 1593.70 | 718.60 | 4169.75 | 4146.23 | 667.57 |
| Leuk3 | 4786.56 | 1532.14 | 1273.42 | 3455.93 | **35.23** | 2092.17 | 849.72 | 4448.63 | 4365.70 | 718.56 |
| 11Tumor | 5859.48 | 1860.34 | 2996.08 | 4257.39 | **249.30** | 2017.90 | 1375.54 | 5015.10 | 5009.23 | 1360.51 |
| Lung2 | 5881.61 | 1826.23 | 2278.62 | 5217.07 | **176.00** | 1987.13 | 1475.88 | 5026.33 | 5010.73 | 1467.35 |
| SMK187 | 8689.50 | 2811.27 | 2996.66 | 5429.74 | **32.93** | 3500.37 | 1359.36 | 7968.70 | 7970.13 | 1320.17 |
| GLI-85 | 9850.18 | 3183.20 | 4602.86 | 7389.20 | **663.67** | 4288.20 | 1136.72 | 8890.40 | 8843.70 | 1111.04 |

HPSO-SSM, SBPSO-D, VS-CCPSO, MFEA, DE, and GA), the numbers of features achieved by MTPSO are the smallest on all datasets, and the classification accuracies are also the highest in almost all datasets. In general, among the ten algorithms, the proposed MTPSO method has the best performance to reduce the number of features and improve the classification accuracy.

### *B. Training Time*

Training time is another important criterion to evaluate the efficiency of the proposed algorithm. Table V records the average training time consumed by these ten approaches on the 30 independent runs. As can be seen from Table V, the training time of MTPSO is the shortest among all the ten algorithms on 11 out of the 18 datasets. Although the training time of MTPSO is a little longer than MFEA on 7 out of 18 datasets, MTPSO achieves better classification accuracies than MFEA on all datasets. This indicates that the proposed knowledge transfer mechanism in MTPSO can effectively implement knowledge transfer between different related tasks to improve the quality of the selected feature subsets. In comparison with the remaining eight algorithms (i.e., CBPSO, PSO(4-2), HPSO-SSM, SBPSO, VLPSO, VS-CCPSO, DE, and GA), their training times are about 2–10 times longer than that of MTPSO. The main reason is that the proposed task generation strategy in MTPSO can effectively generate promising FS tasks with a small number of features, which significantly shortens the computation time of each fitness evaluation during the FS process. In summary, the proposed method can only use a shorter training time to achieve a feature subset with a higher classification accuracy compared with other evolutionary FS methods in most cases.

### *C. Robustness of MTPSO*

Since each algorithm was conducted 30 independent times on the 10-fold cross-validation, it achieved 300 solutions for

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

TABLE V
TRAINING TIME CONSUMED BY THE TEN ALGORITHMS ON THE 18 DATASETS (UNIT: S)

| Datasets | CBPSO | PSO(4-2) | HPSO-SSM | SBPSO-D | VLPSO | VS-CCPSO | MFEA | DE | GA | MTPSO |
|---|---|---|---|---|---|---|---|---|---|---|
| SRBCT | 41.73 | 22.82 | 22.65 | 79.73 | 82.61 | 34.26 | **10.76** | 68.73 | 22.33 | 12.10 |
| PCMAC | 23124.67 | 11070.72 | 16109.56 | 24748.10 | 10689.60 | 28022.07 | 2315.84 | 29730.05 | 19117.17 | **2110.17** |
| Lung1 | 460.93 | 231.01 | 283.03 | 628.03 | 1513.14 | 349.08 | 183.28 | 738.37 | 231.06 | **144.41** |
| RELATHE | 17477.32 | 8756.52 | 12188.67 | 17796.97 | 9755.40 | 21571.24 | 3857.98 | 31647.24 | 18129.39 | **1989.32** |
| BASEHOCK | 37343.45 | 23829.99 | 29413.59 | 39281.09 | 23711.40 | 38586.56 | **5046.22** | 24785.87 | 47602.12 | 5998.02 |
| Leuk1 | 177.22 | 98.07 | 98.18 | 407.96 | 365.37 | 88.80 | **37.64** | 278.09 | 97.20 | 42.99 |
| DLBCL | 203.44 | 109.35 | 94.24 | 422.06 | 430.82 | 89.16 | 48.26 | 342.60 | 113.97 | **38.82** |
| 9Tumor | 147.48 | 89.55 | 122.55 | 380.80 | 338.83 | 69.99 | 56.75 | 227.23 | 78.82 | **42.13** |
| Brain1 | 302.67 | 170.29 | 192.16 | 581.60 | 572.72 | 206.24 | **89.50** | 532.92 | 165.50 | 98.53 |
| Prostate1 | 392.66 | 216.20 | 205.11 | 747.49 | 663.13 | 182.28 | **79.84** | 679.63 | 224.69 | 86.33 |
| Leuk2 | 263.27 | 147.50 | 178.44 | 595.86 | 537.76 | 146.32 | 76.88 | 413.13 | 138.17 | **58.07** |
| Brain2 | 207.95 | 113.70 | 159.05 | 662.88 | 705.81 | 120.37 | 89.62 | 324.09 | 114.36 | **72.13** |
| Prostate2 | 747.82 | 394.83 | 323.44 | 1218.16 | 1353.19 | 376.13 | 187.76 | 1234.16 | 395.57 | **139.44** |
| Leuk3 | 411.10 | 231.77 | 187.31 | 941.55 | 956.63 | 249.38 | **84.11** | 695.22 | 248.23 | 92.11 |
| 11Tumor | 2563.47 | 1331.37 | 1684.06 | 3349.17 | 4044.80 | 1350.91 | 656.48 | 3773.15 | 1243.53 | **644.34** |
| Lung2 | 3452.02 | 1738.49 | 1860.64 | 4447.30 | 4680.03 | 1773.55 | **912.80** | 5491.09 | 1755.84 | 936.33 |
| SMK187 | 4539.39 | 2481.70 | 2309.43 | 6319.76 | 5880.07 | 2608.00 | 779.36 | 7756.35 | 2441.54 | **732.42** |
| GLI-85 | 1192.18 | 611.21 | 777.87 | 2412.91 | 3297.23 | 693.80 | 197.64 | 1952.51 | 661.53 | **178.05** |

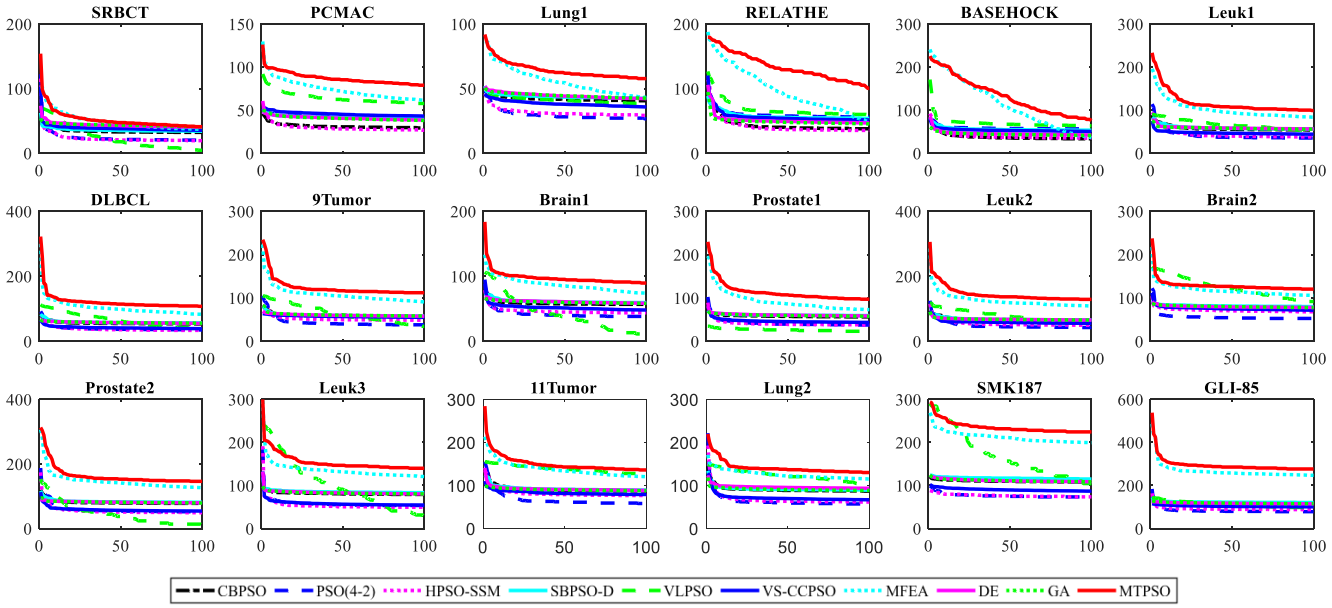



Fig. 7. $Z$-score of the top 100 selected features on the 18 datasets.

each classification dataset. For a dataset, a good algorithm should be one that can identify the crucial features and not by chance. Therefore, we investigate the robustness of MTPSO and the other nine compared algorithms.

In this study, $Z$-score analysis [55] is applied to determine the robustness. $Z$-score is an effective measure to illustrate the significance of the frequency of a feature selected. A feature with a high $Z$-score value indicates that it is not chosen by chance, and an approach that identifies a feature subset containing features with higher $Z$-scores is deemed to be more robust. Equation (12) is used to calculate the $Z$-score of feature $i$

$$Z_i = \frac{f_i - \mu}{\sigma} \tag{12}$$

where $f_i$ represents the number of times feature $i$ being chosen among the $M$ solutions ($M$ is 300 in this $Z$-score analysis). $\mu$ and $\sigma$ indicate the mean and Std of $f_i$, respectively. They are determined as follows:

$$\mu = P(f_i) * M \tag{13}$$

$$\sigma = \sqrt{P(f_i) * (1 - P(f_i)) * M} \tag{14}$$

where $P(f_i)$ represents the probability of feature $i$ being selected from the original feature set, which is equal to the average number of selected features in the $M$ solutions divided by the total number of features.

We sort all features in descending order in terms of their selection frequency. The $Z$-score of the top 100 selected features for MTPSO and the other five approaches on the 18 classification problems are plotted in Fig. 7. The $x$-axis and $y$-axis indicate the sorted top 100 most selected features and the value of $Z$-scores, respectively. From Fig. 7, it can be seen that the features returned by the proposed

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CLASSIFICATION ACCURACIES OF MTPSO AND FIVE NON-EA FS METHODS ON THE 18 DATASETS

| Dataset | $S^2$DFS | EGCFS | MFS-MCDM | MSVM-RFE | RFE-CBR | MTPSO |
|---|---|---|---|---|---|---|
| SRBCT | 92.59(+) | 88.89(+) | **100.00**(≈) | 86.74(+) | 81.48(+) | **100.00** |
| PCMAC | 77.23(+) | 61.47(+) | 78.94(+) | 91.34(−) | **94.01**(−) | 87.84 |
| Lung1 | 84.13(+) | 92.06(+) | 88.89(+) | 72.41(+) | 74.60(+) | **97.40** |
| RELATHE | 70.63(+) | 82.52(≈) | 81.35(+) | 80.26(+) | 76.41(+) | **83.07** |
| BASEHOCK | 84.14(+) | 76.63(+) | 87.81(+) | 86.45(+) | 89.65(+) | **93.29** |
| Leuk1 | 91.30(+) | 73.91(+) | 82.61(+) | 88.85(+) | 87.83(+) | **98.24** |
| DLBCL | 91.67(+) | 79.17(+) | 95.83(+) | 79.99(+) | 83.33(+) | **98.89** |
| 9Tumor | 46.36(+) | 47.94(+) | 51.82(+) | 52.35(+) | 48.18(+) | **69.06** |
| Brain1 | 71.43(+) | 82.14(+) | 78.57(+) | 72.73(+) | 75.00(+) | **89.48** |
| Prostate1 | 80.65(+) | 87.10(+) | 77.42(+) | 91.36(+) | **96.77**(−) | 92.77 |
| Leuk2 | 83.33(+) | 75.00(+) | 85.23(+) | 76.52(+) | 74.17(+) | **92.05** |
| Brain2 | 76.67(+) | 63.33(+) | 74.44(+) | 79.16(+) | 77.78(+) | **85.40** |
| Prostate2 | 84.74(+) | 80.65(+) | 77.42(+) | 85.36(+) | 87.74(+) | **91.77** |
| Leuk3 | 91.30(+) | 95.65(+) | 81.62(+) | 87.08(+) | 89.57(+) | **97.43** |
| 11Tumor | 78.63(+) | 85.86(+) | 84.97(+) | 78.64(+) | 81.58(+) | **91.86** |
| Lung2 | 85.71(+) | 90.48(+) | 90.48(+) | 86.17(+) | 85.71(+) | **96.67** |
| SMK187 | 65.52(+) | 65.52(+) | 68.97(+) | 72.80(+) | 75.86(≈) | **75.96** |
| GLI-85 | 92.31(≈) | 88.46(+) | 88.46(+) | 83.66(+) | 84.62(+) | **92.77** |

MTPSO algorithm have much larger $Z$-score values than other nine compared algorithms on 13 out of 18 datasets (i.e., PCMAC, Lung1, RELATHE, Leuk1, DLBCL, 9Tumor, Brain1, Prostate1, Leuk2, Protate2, 11Tumor, Lung2, and GLI-85). Although the top-ranked features selected by VLPSO on the Brain2, Leuk3, and SMK187 datasets and MFEA on the SRBCT and BASEHOCK datasets has higher $Z$-score than MTPSO, the $Z$-score of the following features in VLPSO and MFEA drops sharply while MTPSO still maintains these values with a flatter slope. In general, features selected by MTPSO are more robust than other counterpart algorithms.

### *D. Comparisons With Non-EA Feature Selection Methods*

To further evaluate the effectiveness of the proposed method, MTPSO is compared with three state-of-the-art non-EA-based FS methods, which are subspace sparsity discriminant FS ($S^2$DFS) [56], FS via adaptive graph learning and constraint (EGCFS) [57], multilabel FS using multicriteria decision making (MFS-MCDM) [58]. These three methods were just published in 2020. In addition, we also compare the proposed method with the support vector machine with recursive feature elimination (SVM-RFE) [59] family methods, which are extensions of SVM-RFE (MSVM-RFE) [60], and SVM-RFE with the correlation bias reduction (RFE-CBR) [61]. Noted that, the parameter setting of these five methods are based on their corresponding references.

Table VI shows the results of MTPSO and the five non-EA FS methods in terms of the classification accuracy. As can be seen from Table VI, the proposed MTPSO method achieves the highest classification accuracy on 16 out of the 18 datasets. RFE-CBR is the second most effective method at obtaining the highest classification accuracy on the PCMAC and Prostate1 datasets. In summary, the proposed FS method outperforms other five non-EA-based FS methods on almost all datasets.

### *E. Analysis on the Two Strategies*

*1) Strategy for Generating Related Task:* To evaluate the effectiveness of the strategy of generating multiple related tasks in the proposed FS method, this experiment analysis the results of MTPSO with this strategy and MTPSO with the random generation strategy, MTPSO$^r$ for short. Table VII records the training time (Time), the average the feature subset size (AvgNF), the best classification accuracy (BeAcc), the average classification accuracy, and its Std based on 30 the independent runs. As can be seen from Table VII, we can see that the proposed MTPSO method achieves the higher average classification accuracies on 17 out of the 18 datasets. In terms of the number of selected features, MTPSO selects fewer features than that of MTPSO$^r$ on all datasets. In terms of the training time, the proposed method is shorter than that of MTPSO$^r$ on 14 out of the 18 datasets. In addition, the results of the significance test indicate that MTPSO obtains significantly better results than MTPSO$^r$ on ten datasets (i.e., RELATHE, Leuk1, 9Tumor, Brain1, Prostate1, Leuk2, Brain2, 11Tumor, Lung2, and SMK187) and similar on the remaining eight datasets (i.e., SRBCT, PCMAC, Lung1, BASEHOCK, DLBCL, Prostate2, Leuk3, and GLI-85). The main reason is that the proposed task generation strategy is developed based on the characteristic of data, which can effectively enhance the quality of the generated FS tasks. Therefore, the strategy for producing related task is effective to enhance the classification performance of the proposed FS approach on the high-dimensional data.

TABLE VII
RESULTS OF USING THE PROPOSED TASK GENERATION STRATEGY AND RANDOM GENERATION STRATEGY ON THE 18 DATASETS

| Datasets | Methods | Time (s) | AvgNF | BeAcc | AvgAcc±Std |
|---|---|---|---|---|---|
| SRBCT | MTPSO$^r$ | 12.52 | 177.58 | 100.00 | 99.32±0.51(≈) |
| | MTPSO | 12.10 | 126.50 | **100.00** | **100.00±0.00** |
| PCMAC | MTPSO$^r$ | 2124.10 | 69.38 | 89.36 | 87.19±0.64(≈) |
| | MTPSO | 2110.17 | 62.47 | **89.61** | **87.84±0.21** |
| Lung1 | MTPSO$^r$ | 141.96 | 353.15 | 98.43 | 97.15±0.58(≈) |
| | MTPSO | 144.41 | 343.24 | **98.50** | **97.40±0.54** |
| RELATHE | MTPSO$^r$ | 2021.80 | 94.17 | 85.77 | 81.85±0.34(+) |
| | MTPSO | 1989.32 | 83.60 | **85.93** | **83.07±0.16** |
| BASEHOCK | MTPSO$^r$ | 5986.17 | 134.24 | 95.54 | 92.98±0.41(≈) |
| | MTPSO | 5998.02 | 118.83 | **95.64** | **93.29±0.28** |
| Leuk1 | MTPSO$^r$ | 44.67 | 376.53 | 100.00 | 97.16±1.28(+) |
| | MTPSO | 42.99 | 317.00 | **100.00** | **98.24±0.81** |
| DLBCL | MTPSO$^r$ | 38.86 | 274.71 | 100.00 | 98.65±1.25(≈) |
| | MTPSO | 38.82 | 237.36 | **100.00** | **98.89±0.81** |
| 9Tumor | MTPSO$^r$ | 41.45 | 432.71 | 66.67 | 61.94±3.25(+) |
| | MTPSO | 42.13 | 423.44 | **73.33** | **69.06±2.34** |
| Brain1 | MTPSO$^r$ | 101.65 | 596.27 | 92.13 | 88.15±1.26(+) |
| | MTPSO | 98.53 | 587.00 | **92.22** | **89.48±1.56** |
| Prostate1 | MTPSO$^r$ | 89.71 | 410.42 | 93.79 | 91.02±1.08(+) |
| | MTPSO | 86.33 | 368.09 | **94.00** | **92.77±0.68** |
| Leuk2 | MTPSO$^r$ | 60.44 | 474.30 | 93.77 | 91.06±1.66(+) |
| | MTPSO | 58.07 | 434.84 | **94.46** | **92.05±0.97** |
| Brain2 | MTPSO$^r$ | 77.83 | 1079.90 | 91.99 | 83.20±1.79(+) |
| | MTPSO | 72.13 | 1066.32 | **92.01** | **85.40±2.47** |
| Prostate2 | MTPSO$^r$ | 146.90 | 712.10 | **94.18** | **91.99±1.46(≈)** |
| | MTPSO | 139.44 | 667.57 | 93.03 | 91.77±0.96 |
| Leuk3 | MTPSO$^r$ | 100.48 | 811.87 | 100.00 | 96.64±1.02(≈) |
| | MTPSO | 92.11 | 718.56 | **100.00** | **97.43±1.12** |
| 11Tumor | MTPSO$^r$ | 675.92 | 1377.76 | 91.44 | 89.69±0.95(+) |
| | MTPSO | 644.34 | 1360.51 | **93.07** | **91.86±1.03** |
| Lung2 | MTPSO$^r$ | 991.40 | 1497.47 | 97.84 | 95.52±0.74(+) |
| | MTPSO | 936.33 | 1467.35 | **98.00** | **96.67±0.52** |
| SMK187 | MTPSO$^r$ | 745.57 | 1342.59 | 76.55 | 73.42±1.71(+) |
| | MTPSO | 732.42 | 1320.17 | **77.51** | **75.96±1.01** |
| GLI-85 | MTPSO$^r$ | 173.22 | 1135.45 | 94.75 | 92.28±1.85(≈) |
| | MTPSO | 178.05 | 1111.04 | **95.00** | **92.77±1.38** |

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
RESULTS OF MTPSO AND OTHER TWO METHOD (MTPSO$^{t-}$ AND MTPSO$^{a}$) ON THE 18 DATASETS

| Datasets | Methods | Time (s) | AvgNF | BeAcc | AvgAcc±Std |
|---|---|---|---|---|---|
| SRBCT | MTPSO$^{t-}$ | 12.88 | 127.57 | 100.00 | 99.16±0.23(≈) |
| | MTPSO$^{a}$ | 12.93 | 129.34 | 100.00 | 99.38±0.19(≈) |
| | MTPSO | 12.10 | 126.50 | **100.00** | **100.00±0.00** |
| PCMAC | MTPSO$^{t-}$ | 2158.74 | 73.57 | 88.62 | 86.89±0.51(≈) |
| | MTPSO$^{a}$ | 2128.05 | 61.87 | 89.37 | 87.54±0.28(≈) |
| | MTPSO | 2110.17 | 62.47 | **89.61** | **87.84±0.21** |
| Lung1 | MTPSO$^{t-}$ | 140.62 | 314.05 | 98.57 | 96.88±0.54(≈) |
| | MTPSO$^{a}$ | 138.92 | 327.38 | **98.66** | 97.34±0.49(≈) |
| | MTPSO | 144.41 | 343.24 | 98.50 | **97.40±0.54** |
| RELATHE | MTPSO$^{t-}$ | 2006.11 | 94.83 | 85.71 | 82.64±0.32(≈) |
| | MTPSO$^{a}$ | 1997.86 | 87.24 | 85.89 | 82.94±0.49(≈) |
| | MTPSO | 1989.32 | 83.60 | **85.93** | **83.07±0.16** |
| BASEHOCK | MTPSO$^{t-}$ | 6018.94 | 141.08 | 95.39 | 91.89±0.31(+) |
| | MTPSO$^{a}$ | 6000.29 | 134.26 | 95.37 | 92.07±0.34(+) |
| | MTPSO | 5998.02 | 118.83 | **95.64** | **93.29±0.28** |
| Leuk1 | MTPSO$^{t-}$ | 40.41 | 307.94 | 100.00 | 96.67±0.98(+) |
| | MTPSO$^{a}$ | 43.22 | 330.85 | 100.00 | 97.97±1.21(≈) |
| | MTPSO | 42.99 | 317.00 | **100.00** | **98.24±0.81** |
| DLBCL | MTPSO$^{t-}$ | 39.86 | 223.36 | 100.00 | 98.51±0.97(≈) |
| | MTPSO$^{a}$ | 40.29 | 231.43 | 100.00 | 98.73±1.10(≈) |
| | MTPSO | 38.82 | 237.36 | **100.00** | **98.89±0.81** |
| 9Tumor | MTPSO$^{t-}$ | 41.65 | 400.50 | 73.13 | 65.17±2.85(+) |
| | MTPSO$^{a}$ | 39.12 | 408.71 | 73.22 | 67.31±2.29(+) |
| | MTPSO | 42.13 | 423.44 | **73.33** | **69.06±2.34** |
| Brain1 | MTPSO$^{t-}$ | 100.83 | 539.60 | 91.11 | 87.52±1.12(+) |
| | MTPSO$^{a}$ | 94.15 | 591.36 | **93.01** | 88.96±1.72(≈) |
| | MTPSO | 98.53 | 587.00 | 92.22 | **89.48±1.56** |
| Prostate1 | MTPSO$^{t-}$ | 83.31 | 304.29 | **94.30** | 90.18±0.90(+) |
| | MTPSO$^{a}$ | 84.19 | 366.29 | 94.16 | 91.69±1.19(+) |
| | MTPSO | 86.33 | 368.09 | 94.00 | **92.77±0.68** |
| Leuk2 | MTPSO$^{t-}$ | 56.42 | 408.27 | 94.39 | 91.75±1.22(≈) |
| | MTPSO$^{a}$ | 59.91 | 439.57 | 94.21 | 91.83±1.27(≈) |
| | MTPSO | 58.07 | 434.84 | **94.46** | **92.05±0.97** |
| Brain2 | MTPSO$^{t-}$ | 75.40 | 967.57 | 91.96 | 84.93±1.86(≈) |
| | MTPSO$^{a}$ | 72.94 | 1047.59 | 92.00 | 85.13±2.31(≈) |
| | MTPSO | 72.13 | 1066.32 | **92.01** | **85.40±2.47** |
| Prostate2 | MTPSO$^{t-}$ | 137.10 | 575.45 | 93.18 | 90.09±1.09(+) |
| | MTPSO$^{a}$ | 146.26 | 671.09 | **93.24** | 91.35±1.67(≈) |
| | MTPSO | 139.44 | 667.57 | 93.03 | **91.77±0.96** |
| Leuk3 | MTPSO$^{t-}$ | 92.13 | 654.10 | 100.00 | 96.00±1.21(+) |
| | MTPSO$^{a}$ | 92.94 | 721.39 | **100.00** | **97.54±1.06(≈)** |
| | MTPSO | 92.11 | 718.56 | 100.00 | 97.43±1.12 |
| 11Tumor | MTPSO$^{t-}$ | 670.83 | 1328.59 | 92.90 | **92.01±0.95**(≈) |
| | MTPSO$^{a}$ | 671.31 | 1367.20 | 92.99 | 91.28±0.89(≈) |
| | MTPSO | 644.34 | 1360.51 | **93.07** | 91.86±1.03 |
| Lung2 | MTPSO$^{t-}$ | 970.00 | 1346.78 | 98.02 | 96.12±0.65(≈) |
| | MTPSO$^{a}$ | 968.47 | 1473.39 | **98.14** | **96.99±0.81(≈)** |
| | MTPSO | 936.33 | 1467.35 | 98.00 | 96.67±0.52 |
| SMK187 | MTPSO$^{t-}$ | 774.98 | 1192.24 | 77.01 | 73.45±1.31(+) |
| | MTPSO$^{a}$ | 770.12 | 1291.59 | 77.27 | 74.29±1.29(+) |
| | MTPSO | 732.42 | 1320.17 | **77.51** | **75.96±1.01** |
| GLI-85 | MTPSO$^{t-}$ | 177.01 | 1005.29 | 94.87 | 92.39±1.74(≈) |
| | MTPSO$^{a}$ | 179.06 | 1139.27 | 94.79 | 92.64±1.61(≈) |
| | MTPSO | 178.05 | 1111.04 | **95.00** | **92.77±1.38** |

*2) Strategy for Implementing Knowledge Transfer:* To investigate the effect of the proposed strategy for implementing knowledge transfer during the FS process, we compare the results of MTPSO with other two methods. The first method is not to use the proposed knowledge transfer strategy during the evolutionary process, which is denoted as MTPSO$^{t-}$. The purpose of the first method is to illustrate the importance of knowledge transfer in the search process. The second method uses the knowledge transfer strategy proposed in [62] for sharing information between different tasks, which is denoted as MTPSO$^{a}$. In this study, the second method aims to illustrate the effectiveness of the proposed knowledge transfer strategy.

Table VIII lists the training time, classification accuracy, and the number of the selected features obtained by MTPSO, MTPSO$^{t-}$, and MTPSO$^{a}$. As can be seen from Table VIII, we can notice that the method with the proposed knowledge transfer strategy achieves better average classification accuracies with smaller feature subsets in shorter training time on most datasets. Compared with MTPSO$^{t-}$, MTPSO achieves significantly better results on eight datasets (i.e., BASEHOCK, Leuk1, 9Tumor, Brain1, Prostate1, Prostate2, Leuk3, and SMK187) and similar on ten datasets (i.e., SRBCT, PCMAC, Lung1, RELATHE, DLBCL, Leuk2, Brain2, 11Tumor, Lung2, and GLI-85). This shows that knowledge transfer plays an important role in the search process of the proposed FS approach. Compared with MTPSO$^{a}$, MTPSO achieves significantly better results on four datasets (i.e., BASEHOCK, 9Tumor, Prostate1, and SMK187) and similar on the remaining fourteen datasets (i.e., SRBCT, PCMAC, Lung1, RELATHE, Leuk1, DLBCL, Brain1, Leuk2, Brain2, Prostate2, Leuk3, 11Tumor, Lung2, and GLI-85). This shows that the proposed knowledge transfer strategy can effectively achieve positive knowledge transfer between tasks by generating new guiding positions. In summary, the proposed knowledge transfer strategy is effective to enhance the performance of MTPSO for solving high-dimensional FS problems.

## VI. CONCLUSION

The goal of this article was to develop an effective FS method for high-dimensional classification problems. The goal has been successfully achieved by proposing an effective multitasking PSO approach for FS that can effectively select a good feature subset through knowledge transfer among related tasks.

The effectiveness and the efficiency of the proposed method are evaluated by comparing it with other state-of-the-art methods in terms of the training time, the classification accuracy, and the number of selected features. The experimental results showed that the proposed MTPSO method can effectively evolve a good feature subset with a higher classification accuracy than its counterparts on almost all datasets. This is because the knowledge transfer strategy in MTPSO can effectively help PSO jump out of potential local optima and evolve better solutions. In terms of the training time, the proposed method uses much less than other compared methods, since the proposed task generation strategy can transform a high-dimensional FS task into several low-dimensional FS tasks. In general, the proposed method can successfully enhance the efficiency of PSO-based FS methods, and achieve a good feature subset for high-dimensional classification problems.

Some interesting directions can be further investigated in the near future. First, how to decide the number of tasks effectively and efficiently by considering the characteristics of the data itself. Second, how to design an effective knowledge transfer mechanism by considering the relevance of features to further improve the performance of the proposed method. Third, how to combine domain expertise (e.g., the gene expression domain) to further reduce the number of selected features to improve the efficiency of validating

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

selected features in practice. In addition, treating FS as a multitasking multiobjective optimization problem is a promising research direction, which can provide decision-makers with more tradeoff solutions.

## REFERENCES


[1] R. O. Duda, P. E. Hart, and D. G. Stork, *Pattern Classification*. New York, NY, USA: Wiley, 2012.

[2] F. Zhang, Y. Mei, S. Nguyen, and M. Zhang, "Evolving scheduling heuristics via genetic programming with feature selection in dynamic flexible job shop scheduling," *IEEE Trans. Cybern.*, vol. 51, no. 4, pp. 1797–1811, Apr. 2021.

[3] R. A. Sinoara, J. Camacho-Collados, R. G. Rossi, R. Navigli, and S. O. Rezende, "Knowledge-enhanced document embeddings for text classification," *Knowl. Based Syst.*, vol. 163, pp. 955–971, Jan. 2019.

[4] Z. Yu *et al.*, "Adaptive semi-supervised classifier ensemble for high dimensional data classification," *IEEE Trans. Cybern.*, vol. 49, no. 2, pp. 366–379, Feb. 2019.

[5] C. Wang, Q. Hu, X. Wang, D. Chen, Y. Qian, and Z. Dong, "Feature selection based on neighborhood discrimination index," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 7, pp. 2986–2999, Jul. 2018.

[6] B. Xue, M. Zhang, and W. N. Browne, "Particle swarm optimisation for feature selection in classification: Novel initialisation and updating mechanisms," *Appl. Soft Comput.*, vol. 18, pp. 261–276, May 2014.

[7] B. Tran, B. Xue, and M. Zhang, "A new representation in PSO for discretization-based feature selection," *IEEE Trans. Cybern.*, vol. 48, no. 6, pp. 1733–1746, Jun. 2018.

[8] K. Chen, F. Zhou, and B. Xue, "Particle swarm optimization for feature selection with adaptive mechanism and new updating strategy," in *Proc. Aust. Joint Conf. Artif. Intell.*, 2018, pp. 419–431.

[9] G. I. Sayed, A. E. Hassanien, and A. T. Azar, "Feature selection via a novel chaotic crow search algorithm," *Neural Comput. Appl.*, vol. 31, no. 1, pp. 171–188, 2019.

[10] J. Tang, S. Alelyani, and H. Liu, "Feature selection for classification: A review," in *Data Classification: Algorithms and Applications*. Boca Raton, Fl, USA: Taylor Francis, 2014, p. 37.

[11] Y. Zhang, H. Li, Q. Wang, and C. Peng, "A filter-based bare-bone particle swarm optimization algorithm for unsupervised feature selection," *Appl. Intell.*, vol. 49, no. 8, pp. 2889–2898, 2019.

[12] S. Singha and P. P. Shenoy, "An adaptive heuristic for feature selection based on complementarity," *Mach. Learn.*, vol. 107, no. 12, pp. 2027–2071, 2018.

[13] K.-L. Du and M. Swamy, "Particle swarm optimization," in *Search and Optimization by Metaheuristics*. Cham, Switzerland: Springer, 2016, pp. 153–173.

[14] K. Chen, B. Xue, M. Zhang, and F. Zhou, "Novel chaotic grouping particle swarm optimization with a dynamic regrouping strategy for solving numerical optimization tasks," *Knowl. Based Syst.*, vol. 194, Apr. 2020, Art. no. 105568.

[15] B. Xue, M. Zhang, W. N. Browne, and X. Yao, "A survey on evolutionary computation approaches to feature selection," *IEEE Trans. Evol. Comput.*, vol. 20, no. 4, pp. 606–626, Aug. 2016.

[16] X. Song, Y. Zhang, Y.-N. Guo, X.-Y. Sun, and Y.-L. Wang, "Variable-size cooperative coevolutionary particle swarm optimization for feature selection on high-dimensional data," *IEEE Trans. Evol. Comput.*, vol. 24, no. 5, pp. 882–895, Oct. 2020.

[17] B. H. Nguyen, B. Xue, and M. Zhang, "A survey on swarm intelligence approaches to feature selection in data mining," *Swarm Evol. Comput.*, vol. 54, May 2020, Art. no. 100663.

[18] Y. Hu, Y. Zhang, and D. Gong, "Multiobjective particle swarm optimization for feature selection with fuzzy cost," *IEEE Trans. Cybern.*, vol. 51, no. 2, pp. 874–888, Feb. 2021.

[19] B. Tran, B. Xue, and M. Zhang, "Variable-length particle swarm optimization for feature selection on high-dimensional classification," *IEEE Trans. Evol. Comput.*, vol. 23, no. 3, pp. 473–487, Jun. 2019.

[20] A. Gupta, Y.-S. Ong, and L. Feng, "Multifactorial evolution: Toward evolutionary multitasking," *IEEE Trans. Evol. Comput.*, vol. 20, no. 3, pp. 343–357, Jun. 2016.

[21] A. Gupta, Y.-S. Ong, L. Feng, and K. C. Tan, "Multiobjective multifactorial optimization in evolutionary multitasking," *IEEE Trans. Cybern.*, vol. 47, no. 7, pp. 1652–1665, Jul. 2017.

[22] Y. Yuan, Y.-S. Ong, A. Gupta, P. S. Tan, and H. Xu, "Evolutionary multitasking in permutation-based combinatorial optimization problems: Realization with TSP, QAP, LOP, and JSP," in *Proc. IEEE Reg. 10 Conf.*, 2016, pp. 3157–3164.

[23] F. Zhang, Y. Mei, S. Nguyen, and M. Zhang, "A preliminary approach to evolutionary multitasking for dynamic flexible job shop scheduling via genetic programming," in *Proc. Genet. Evol. Comput. Conf. Compan.*, 2020, pp. 107–108.

[24] J. Ding, C. Yang, Y. Jin, and T. Chai, "Generalized multitasking for evolutionary optimization of expensive problems," *IEEE Trans. Evol. Comput.*, vol. 23, no. 1, pp. 44–58, Feb. 2019.

[25] L. Feng *et al.*, "Explicit evolutionary multitasking for combinatorial optimization: A case study on capacitated vehicle routing problem," *IEEE Trans. Cybern.*, vol. 51, no. 6, pp. 3143–3156, Jun. 2021.

[26] L. Zhou *et al.*, "Toward adaptive knowledge transfer in multifactorial evolutionary computation," *IEEE Trans. Cybern.*, vol. 51, no. 5, pp. 2563–2576, May 2021.

[27] B. Da *et al.*, "Evolutionary multitasking for single-objective continuous optimization: Benchmark problems, performance metric, and baseline results," 2017. [Online]. Available: arXiv:1706.03470.

[28] A. Gupta, Y.-S. Ong, B. Da, L. Feng, and S. D. Handoko, "Landscape synergy in evolutionary multitasking," in *Proc. IEEE Congr. Evol. Comput.*, 2016, pp. 3076–3083.

[29] F. Zhang, Y. Mei, S. Nguyen, M. Zhang, and K. C. Tan, "Surrogate-assisted evolutionary multitask genetic programming for dynamic flexible job shop scheduling," *IEEE Trans. Evol. Comput.*, early access, Mar. 12, 2021, doi: 10.1109/TEVC.2021.3065707.

[30] K. Chen, B. Xue, M. Zhang, and F. Zhou, "An evolutionary multitasking-based feature selection method for high-dimensional classification," *IEEE Trans. Cybern.*, early access, Dec. 31, 2020, doi: 10.1109/TCYB.2020.3042243.

[31] Y.-S. Ong and A. Gupta, "Evolutionary multitasking: A computer science view of cognitive multitasking," *Cogn. Comput.*, vol. 8, no. 2, pp. 125–142, 2016.

[32] J. Yi, J. Bai, H. He, W. Zhou, and L. Yao, "A multifactorial evolutionary algorithm for multitasking under interval uncertainties," *IEEE Trans. Evol. Comput.*, vol. 24, no. 5, pp. 908–922, Oct. 2020.

[33] L. Zhou, L. Feng, J. Zhong, Y.-S. Ong, Z. Zhu, and E. Sha, "Evolutionary multitasking in combinatorial search spaces: A case study in capacitated vehicle routing problem," in *Proc. IEEE Symp. Series Comput. Intell.*, 2016, pp. 1–8.

[34] G. Li, Q. Lin, and W. Gao, "Multifactorial optimization via explicit multipopulation evolutionary framework," *Inf. Sci.*, vol. 512, pp. 1555–1570, Feb. 2020.

[35] J. Zhong, L. Feng, W. Cai, and Y.-S. Ong, "Multifactorial genetic programming for symbolic regression problems," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 11, pp. 4492–4505, Nov. 2020.

[36] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proc. Int. Conf. Neural Netw.*, vol. 4, 1995, pp. 1942–1948.

[37] I. Jain, V. K. Jain, and R. Jain, "Correlation feature selection based improved-Binary Particle Swarm Optimization for gene selection and cancer classification," *Appl. Soft Comput.*, vol. 62, pp. 203–215, Jan. 2018.

[38] B. Tran, B. Xue, M. Zhang, and S. Nguyen, "Investigation on particle swarm optimisation for feature selection on high-dimensional data: Local search and selection bias," *Connect. Sci.*, vol. 28, no. 3, pp. 270–294, 2016.

[39] M. Robnik-Šikonja and I. Kononenko, "Theoretical and empirical analysis of relieff and RReliefF," *Mach. Learn.*, vol. 53, nos. 1–2, pp. 23–69, 2003.

[40] I. Das, "On characterizing the 'knee' of the pareto curve based on normal-boundary intersection," *Struct. Optim.*, vol. 18, nos. 2–3, pp. 107–115, 1999.

[41] M. Jiang, Z. Wang, H. Hong, and G. G. Yen, "Knee point based imbalanced transfer learning for dynamic multi-objective optimization," *IEEE Trans. Evol. Comput.*, vol. 25, no. 1, pp. 117–129, Feb. 2021.

[42] Z. Tang, M. Gong, F. Jiang, H. Li, and Y. Wu, "Multipopulation optimization for multitask optimization," in *Proc. IEEE Congr. Evol. Comput.*, 2019, pp. 1906–1913.

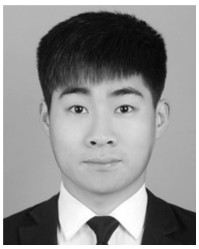
[43] F. Zhang, Y. Mei, S. Nguyen, K. C. Tan, and M. Zhang, "Multitask genetic programming-based generative hyperheuristics: A case study in dynamic scheduling," *IEEE Trans. Cybern.*, early access, Mar. 22, 2021, doi: 10.1109/TCYB.2021.3065340.

[44] T. Baeck, D. Fogel, and Z. Michalewicz, "Tournament selection," in *Evolutionary Computation*. Boca Raton, FL, USA: CRC Press, 2018, pp. 219–224.

[45] T. Yalcinoz, H. Altun, and M. Uzam, "Economic dispatch solution using a genetic algorithm based on arithmetic crossover," in *Proc. IEEE Porto Power Tech*, vol. 2, 2001, pp. 1–4.


Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.

[46] K. Chen, F.-Y. Zhou, and X.-F. Yuan, "Hybrid particle swarm optimization with spiral-shaped mechanism for feature selection," *Expert Syst. Appl.*, vol. 128, pp. 140–156, Aug. 2019.
[47] G. Patterson and M. Zhang, "Fitness functions in genetic programming for classification with unbalanced data," in *Proc. Aust. Joint Conf. Artif. Intell.*, 2007, pp. 769–775.
[48] L.-Y. Chuang, C.-H. Yang, and J.-C. Li, "Chaotic maps based on binary particle swarm optimization for feature selection," *Appl. Soft Comput.*, vol. 11, no. 1, pp. 239–248, 2011.
[49] B. H. Nguyen, B. Xue, P. Andreae, and M. Zhang, "A new binary particle swarm optimization approach: Momentum and dynamic balance between exploration and exploitation," *IEEE Trans. Cybern.*, vol. 51, no. 2, pp. 589–603, Feb. 2021.
[50] R. N. Khushaba, A. Al-Ani, and A. Al-Jumaily, "Differential evolution based feature subset selection," in *Proc. Int. Conf. Pattern Recognit.*, 2008, pp. 1–4.
[51] F. Tan, X. Fu, Y. Zhang, and A. G. Bourgeois, "A genetic algorithm-based method for feature subset selection," *Soft Comput.*, vol. 12, no. 2, pp. 111–120, 2008.
[52] H. B. Nguyen, B. Xue, I. Liu, P. Andreae, and M. Zhang, "New mechanism for archive maintenance in PSO-based multi-objective feature selection," *Soft Comput.*, vol. 20, no. 10, pp. 3927–3946, 2016.
[53] P. Cunningham and S. J. Delany, "K-nearest neighbour classifiers," *Multiple Classifier Syst.*, vol. 34, no. 8, pp. 1–17, 2007.
[54] H. B. Nguyen, B. Xue, and P. Andreae, "PSO with surrogate models for feature selection: Static and dynamic clustering-based methods," *Memetic Comput.*, vol. 10, no. 3, pp. 291–300, 2018.
[55] Z. Zhu, Y.-S. Ong, and M. Dash, "Markov blanket-embedded genetic algorithm for gene selection," *Pattern Recognit.*, vol. 40, no. 11, pp. 3236–3248, 2007.
[56] Z. Wang, F. Nie, L. Tian, R. Wang, and X. Li, "Discriminative feature selection via a structured sparse subspace learning module," in *Proc. Int. Joint Conf. Artif. Intell.*, 2020, pp. 3009–3015.
[57] R. Zhang, Y. Zhang, and X. Li, "Unsupervised feature selection via adaptive graph learning and constraint," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Dec. 24, 2020, doi: 10.1109/TNNLS.2020.3042330.
[58] A. Hashemi, M. B. Dowlatshahi, and H. Nezamabadi-pour, "MFS-MCDM: Multi-label feature selection using multi-criteria decision making," *Knowl. Based Syst.*, vol. 206, Oct. 2020, Art. no. 106365.
[59] I. Guyon, J. Weston, S. Barnhill, and V. Vapnik, "Gene selection for cancer classification using support vector machines," *Mach. Learn.*, vol. 46, no. 1, pp. 389–422, 2002.
[60] X. Zhou and D. P. Tuck, "MSVM-RFE: Extensions of SVM-RFE for multiclass gene selection on DNA microarray data," *Bioinformatics*, vol. 23, no. 9, pp. 1106–1114, 2007.
[61] K. Yan and D. Zhang, "Feature selection and analysis on correlated gas sensor data with recursive feature elimination," *Sens. Actuators, B Chem.*, vol. 212, pp. 353–363, Jun. 2015.
[62] L. Feng *et al.*, "An empirical study of multifactorial PSO and multifactorial DE," in *Proc. IEEE Congr. Evol. Comput.*, 2017, pp. 921–928.



**Ke Chen** (Member, IEEE) received the B.Sc. degree from Weifang University, Weifang, China, in 2014, the M.Sc. degree from Yanshan University, Qinhuangdao, China, in 2017, and the Ph.D. degree from Shandong University, Jinan, China, in 2021.

He is currently a Research Associate Professor with the School of Electrical Engineering, Zhengzhou University, Zhengzhou, China. He has over 20 journal and conference papers. His current research interests include evolutionary computation, feature selection, and multitasking.



**Bing Xue** (Senior Member, IEEE) received the B.Sc. degree from the Henan University of Economics and Law, Zhengzhou, China, in 2007, the M.Sc. degree in management from Shenzhen University, Shenzhen, China, in 2010, and the Ph.D. degree in computer science from the Victoria University of Wellington (VUW), Wellington, New Zealand, in 2014.

She is currently a Professor of Computer Science, and the Program Director of Science with the School of Engineering and Computer Science, VUW. She has over 300 papers published in fully refereed international journals and conferences. Her research focuses mainly on evolutionary computation and machine learning.

Dr. Xue is currently the Chair of IEEE Computational Intelligence Society (CIS) Task Force on Transfer Learning & Transfer Optimization, the Vice-Chair of IEEE CIS Evolutionary Computation Technical Committee, and an Editor of IEEE CIS Newsletter. She is also served as an Associate Editor of several international journals, such as *IEEE Computational Intelligence Magazine* and IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION.



**Mengjie Zhang** (Fellow, IEEE) received the B.E. and M.E. degrees from the Artificial Intelligence Research Center, Agricultural University of Hebei, Hebei, China, in 1989 and 1992, respectively, and the Ph.D. degree in computer science from RMIT University, Melbourne, VIC, Australia, in 2000.

He is currently a Professor of Computer Science, the Head of the Evolutionary Computation Research Group, and the Associate Dean (Research and Innovation) with the Faculty of Engineering, Victoria University of Wellington, Wellington, New Zealand. He has published over 600 research papers in refereed international journals and conferences. His current research interest includes evolutionary computation.

Prof. Zhang was the Chair of the IEEE CIS Intelligent Systems and Applications Technical Committee and the IEEE CIS Emergent Technologies Technical Committee and the Evolutionary Computation Technical Committee, and a member of the IEEE CIS Award Committee. He is a Fellow of Royal Society of New Zealand and he has been a Panel Member of the Marsden Fund (New Zealand Government Funding), and a member of ACM.



**Fengyu Zhou** received the B.E. and M.E. degrees from the Shandong University of Science and Technology, Jinan, China, in 1992 and 1996, and the Ph.D. degree in electrical engineering from Tianjin University, Tianjin, China, in 2008.

He is currently a Professor with School of the Control Science and Engineering, Shandong University, Jinan. He has published more than 150 academic papers in international journals and conferences. His current research interests include evolutionary computation, and intelligent robot.

Prof. Zhou is the Chief Scientist of National Key Research and Development Projects, a Senior Member of China Automation Society, a member of Construction Robot Professional Committee, and the Director of Science Popularization and Innovation Working Committee of Shandong Automation Society.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 24,2026 at 06:46:37 UTC from IEEE Xplore. Restrictions apply.